

Eyes on Target: What Neurons Must do for the Vestibuloocular Reflex During Linear Motion

Dora E. Angelaki

Department of Neurobiology, Washington University School of Medicine, St. Louis, Missouri 63110

Submitted 14 January 2004; accepted in final form 24 February 2004

Angelaki, Dora E. Eyes on target: what neurons must do for the vestibuloocular reflex during linear motion. *J Neurophysiol* 92: 20–35, 2004; 10.1152/jn.00047.2004. A gaze-stabilization reflex that has been conserved throughout evolution is the rotational vestibuloocular reflex (RVOR), which keeps images stable on the entire retina during head rotation. An ethological newer reflex, the translational or linear VOR (TVOR), provides fast foveal image stabilization during linear motion. Whereas the sensorimotor processing has been extensively studied in the RVOR, much less is currently known about the neural organization of the TVOR. Here we summarize the computational problems faced by the system and the potential solutions that might be used by brain stem and cerebellar neurons participating in the VORs. First and foremost, recent experimental and theoretical evidence has shown that, contrary to popular beliefs, the sensory signals driving the TVOR arise from both the otolith organs and the semicircular canals. Additional unresolved issues include a scaling by both eye position and vergence angle as well as the temporal transformation of linear acceleration signals into eye-position commands. Behavioral differences between the RVOR and TVOR, as well as distinct differences in neuroanatomical and neurophysiological properties, raise multiple functional questions and computational issues, only some of which are readily understood. In this review, we provide a summary of what is known about the functional properties and neural substrates for this oculomotor system and outline some specific hypotheses about how sensory information is centrally processed to create motor commands for the VORs.

INTRODUCTION

The world is viewed from a constantly shifting platform, where visual mechanisms function optimally only if images on the retina remain stable. This is achieved primarily by the vestibuloocular reflexes (VORs), which provide fast compensation during both translational and rotational motions. Although the rotational VOR (RVOR) is phylogenetically older and highly conserved throughout evolution, the translational VOR (TVOR) represents a relatively recent evolutionary acquisition that appears to have evolved in parallel with foveal vision, vergence eye movements, and stereopsis (Miles 1993, 1998). Its amplitude and functional properties are only well described for humans and non-human primates (Angelaki and McHenry 1999; McHenry and Angelaki 2000; Paige 1989; Paige and Tomko 1991a,b; Ramat and Zee 2003; Schwarz and Miles 1991; Schwarz et al. 1989; Telford et al. 1997; Zhou et al. 2003). In contrast, an image-stabilization system appropriate to compensate for the visual consequences of translational motion remains rudimentary or absent in lateral-eyed species (Baarsma and Collewijn 1975; Dickman and Angelaki 1999;

Hess and Dieringer 1990, 1991; Hess et al. 1984; Maruta et al. 2001). The functional goal of the TVOR is to decrease conjugate retinal slip and minimize binocular disparities during either self-motion or passive displacements such as those experienced when riding a car (Hess and Angelaki 2003b; McHenry and Angelaki 2000). These vestibular-driven eye movements, which are vital for visual acuity, complement and work closely in synergy with visuomotor reflexes (e.g., ocular following, OFR), which sense the subject's motion by decoding either the pattern of optic flow or depth and binocular-disparity cues (Busettini et al. 1994, 1996a,b, 1997; Miles and Busettini 1992; Miles et al. 1991; Schwarz et al. 1989; Yang et al. 1999).

Unlike the RVOR where an eye movement that is opposite to head velocity can result in stabilization of the entire visual world, a single eye movement can not typically stabilize images on the whole retina during translation. A functional compromise would then have to place a special emphasis on visual acuity on the fovea. Indeed, compensatory eye movements during translation exhibit a dependence on the direction of gaze as dictated by the geometrical requirements to keep images stable on the two foveae (Angelaki and Hess 2001; McHenry and Angelaki 2000; Seidman et al. 1999; Tomko and Paige 1992). In addition, both the human and monkey TVORs are proportional to vergence angle (Angelaki 2002; Angelaki and McHenry 1999; Angelaki et al. 2000a; Busettini et al. 1994; Crane et al. 2003; Gianna et al. 1995, 1997, 2000; McHenry and Angelaki 2000; Medendorp et al. 2002; Paige 1989; Paige and Tomko 1991a,b; Paige et al. 1998; Ramat and Zee 2003; Schwarz and Miles 1991; Schwarz et al. 1989; Telford et al. 1997; Zhou et al. 2003). Such a scaling of the TVOR by viewing distance is necessary to compensate for the motion parallax typically experienced during translation.

These unique properties of the TVOR raise multiple questions and computational issues, only some of which have been addressed in recent experiments. In this review, I provide a summary of what is known about the functional properties and neural substrates for this oculomotor subsystem and outline some specific hypotheses about how sensory information is centrally processed to create motor commands for the TVOR. Because the TVOR works in synergy with the OFR and shares many common functional properties, some aspects of the visuo-motor subcortical processing will be also briefly discussed. Because comprehensive reviews exist for the latter (Kawano 1999; Miles 1993, 1995, 1997, 1998; Takemura and Kawano 2002), the focus here will be on the vestibularly

Address: D. Angelaki, Dept. of Neurobiology, Washington University School of Medicine, 660 S. Euclid Ave, St. Louis, MO 63110 (E-mail: angelaki@pcg.wustl.edu).

The costs of publication of this article were defrayed in part by the payment of page charges. The article must therefore be hereby marked "advertisement" in accordance with 18 U.S.C. Section 1734 solely to indicate this fact.

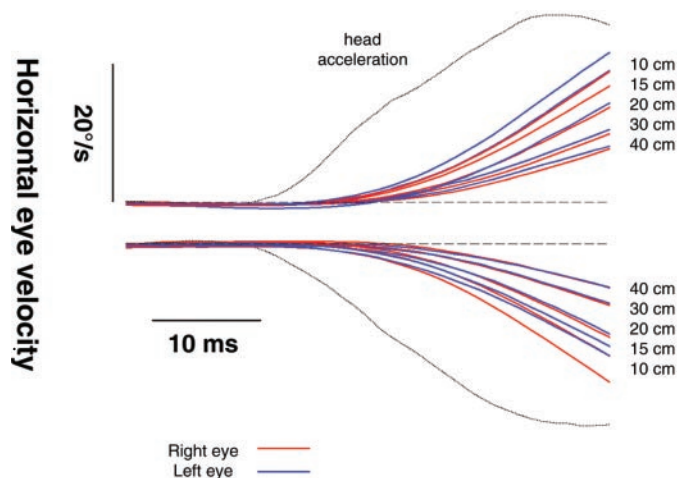


FIG. 1. The translational vestibuloocular reflex (TVOR) depends on viewing distance with the shortest latency. Data plot mean horizontal position of the right and left eyes during rightward and leftward motion (top and bottom traces, respectively) at different viewing distances. The stimulus linear acceleration is shown with black lines (peak of 0.5 *g*). Motion was in complete darkness. Modified from Angelaki and McHenry (1999).

driven eye movements. First, the functional properties of the TVOR will be considered and its unique differences from the RVOR summarized. Next, I will discuss how different neuronal classes, which have been shown to participate in the RVOR, modulate during the TVOR. Finally, the question of how sensory vestibular signals need to be processed to generate the complex patterns of eye movements observed in the TVOR will be considered. Some of the sensorimotor transformation steps are well understood, whereas others are still in their infancy. I will summarize the known facts and then speculate on potential solutions and hypotheses for the missing links to the transformation from sensory vestibular information to motor action.

Different functional goals and distinct functional properties for ocular compensation during rotation and translation

It has become increasingly clear that, although the RVOR and TVOR both originate from motion signals detected by the vestibular labyrinth, there are large differences in both the functional goals and the properties of these eye movements. These differences are summarized in the following paragraphs.

DEPENDENCE ON VIEWING DISTANCE. As already mentioned, the TVOR depends on viewing distance (Paige and Tomko 1991b; Schwarz and Miles 1991), and this dependence is seen at the shortest latency (Fig. 1) (Angelaki and McHenry 1999; Gianna et al. 1995; Zhou et al. 2003). Whether and how much the RVOR exhibits a small dependence on viewing distance has been rather controversial (Chen-Huang and McCrea 1999a,b; Crane and Demer 1998; Paige et al. 1998; Snyder and King 1992). Because the eyes are located forward relative to the vestibular apparatus, a small dependence of semicircular canal-ocular signals on viewing distance would be necessary to account for the equally small translation of the eyes relative to the target during rotation (Hine and Thorn 1987; Viirre et al. 1986). Yet, such an RVOR dependence on viewing distance, even when present, is more than an order of magnitude lower than that of the TVOR.

DEPENDENCE ON GAZE DIRECTION. Compensatory eye movements during translation exhibit a dependence on eye position, as dictated by the geometrical requirements to keep images stable on the fovea (Angelaki and Hess 2001; Hess and Angelaki 2003b; McHenry and Angelaki 2000; Paige and Tomko 1991b; Seidman et al. 1999; Tomko and Paige 1992). Specifically, the horizontal velocity component for each eye, $\dot{\theta}$, during translation with velocity v along a direction forming an angle α with the forward axis of the head should depend on eye-to-target distance, d , and eye position, θ , as follows (Angelaki and Hess 2001)

$$\dot{\theta} = \frac{v}{d} \sin(\alpha - \theta) \quad (1)$$

Equation 1 states that, to keep images stable on the fovea during translation, each eye should rotate with a velocity that increases in inverse proportion of eye-to-target distance, d , and should have an amplitude that is dependent on the difference between gaze and heading directions. For sinusoidal fore-aft motion ($\alpha = 0$), for example, where the dependence on gaze is largest, the eye velocity exhibits a V-shape curve and a reversal in phase as a function of eye position (Fig. 2). For heading 10° to the right and left from straight-ahead ($\alpha = \pm 10^\circ$), the V-shape curves shift to the right and left, respectively, as expected from an eye velocity dependence on the difference ($\alpha - \theta$) in Eq. 1.

FOVEAL RATHER THAN FULL-FIELD IMAGE STABILIZATION. In contrast to the RVOR that can stabilize the whole visual field on the retina, a single eye movement can not typically stabilize images on the whole retina during translation. This property can best be appreciated when considering the radial optic flow experienced during forward motion. In this case, the moving

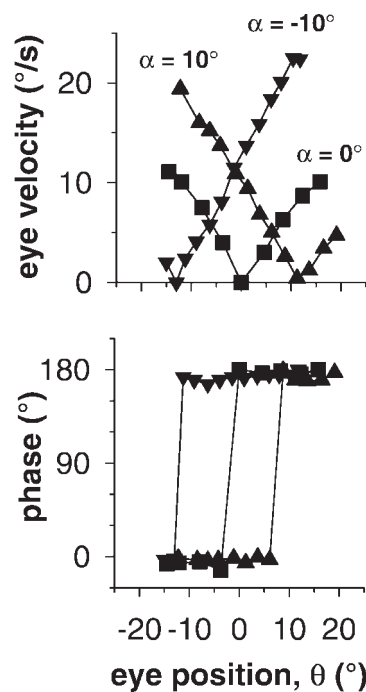


FIG. 2. Eye-position dependence of the TVOR during three heading directions: fore aft ($\alpha = 0^\circ$) and $\pm 10^\circ$ heading from that axis. Horizontal eye velocity follows the eye-position dependence of Eq. 1, i.e., it is of minimal magnitude when $\alpha = \theta$ and increases for larger angles.

subject views an expanding world where objects at the periphery of the visual field move much faster than those in more central portions of the retina. In addition, objects will appear to move in different directions depending on their spatial location within the visual field. Thus *at any given point in time, only images in one particular spatial location in the visual field can remain stable*. It has been shown that the amplitude and direction of the eye movement elicited by the TVOR is appropriate to anticipate and compensate for the resulting optic flow as necessary to minimize image slip on the fovea (according to Eq. 1) (Angelaki and Hess 2001; Hess and Angelaki 2003b; Paige and Tomko 1991b).

THREE-DIMENSIONAL OCULAR KINEMATICS. It has also been shown that the three-dimensional (3D) kinematic properties of the TVOR are different from those of the RVOR and closer to the properties of smooth pursuit eye movements (Angelaki et al. 2000b, 2003). To appreciate the reason behind these differences, let's briefly review the functional requirements regarding ocular torsion. For a reflex the goal of which is to stabilize images on the whole retina, like the RVOR, all degrees of freedom of the eyes are unambiguously specified (Misslisch and Hess 2000; Misslisch et al. 1994). Accordingly, during a rotation of the head about an arbitrary axis, the eyes will rotate not just vertically and horizontally but also in the torsional direction as required by the functional need to have an eye rotation axis that is parallel (but opposite) to the axis of head rotation. For a *foveal* eye-movement system, on the other hand (e.g., smooth pursuit), only the horizontal and vertical components of eye rotation that define gaze direction suffice, whereas ocular torsion remains unspecified. Among all the possible 3D orientations that the eye could use, it is now well established that the third degree of freedom of the eyes during smooth pursuit and all visually guided eye movements obey Listing's law (Haslwanter et al. 1991; Tweed and Vilis 1987, 1990; Tweed et al. 1992). The TVOR and OFR, unlike the RVOR, are *foveal stabilization reflexes*. Thus it is not surprising that both the TVOR (Angelaki et al. 2000b, 2003) and the OFR (B. Adeyemo and D. E. Angelaki, unpublished observations) follow properties similar to pursuit and saccades and not those of the RVOR.

RESPONSE LATENCY. In the monkey, the TVOR has been shown to have a short latency (10–12 ms), thus providing fast compensation for retinal slip (Angelaki and McHenry 1999; Bush and Miles 1996; Zhou et al. 2003). Human TVOR latencies have been reported to be longer (Aw et al. 2003; Crane et al. 2003; Gianna et al. 1995; but see Ramat and Zee 2003), although it is unclear whether they reflect species or experimental differences. There are several technical difficulties that might have hindered a demonstration of shorter latencies for the TVOR in humans. These include acceleration limitations of linear sleds as well as the relative small amplitude of eye movements during (particularly fore-aft) translation. Yet even considering these limitations, it is possible that the TVOR has a longer latency than the 6 to 7-ms latency of the RVOR (Huterer and Cullen 2002; Minor et al. 1999).

Historical perspective

It is important to point out that these properties regarding the existence of a specialized gaze stabilization system during self-motion have only recently been appreciated, following the

pioneering studies of Miles (Schwarz and Miles 1991; Schwarz et al. 1989) and Paige (Paige 1989; Paige and Tomko 1991a,b). Before this time, human eye movements in response to linear acceleration focused on ocular torsion (Collewijn et al. 1985; Diamond et al. 1979; Hannen et al. 1966; Kellogg 1965; Lichtenberg et al. 1982). Accordingly, most of the prior research focused on a different otolith-ocular reflex, typically known as a "tilt" response, that functions to maintain the orientation of the eyes in space during head tilts, thus providing the tonic drive for sustained ocular counterrolling (Paige and Tomko 1991a). This ocular counterrolling is the primary otolith-ocular reflex in all lateral-eyed species and is elicited during either translation or head tilt (Baarsma and Collewijn 1975; Dickman and Angelaki 1999; Hess and Dieringer 1990, 1991). The function to use a gravity-sensing mechanism to maintain gaze relative to the horizon is very important for these animals typically lacking a well-developed saccadic system. In contrast, the otolith-driven ocular counterrolling in primates and humans has a very small gain (10%) (Collewijn et al. 1985; Krejcova et al. 1971) and is maintained through a yet unknown otolith influence on the oculomotor neural integrator (Crawford et al. 2003; Glasauer et al. 2001; Haslwanter et al. 1992; Hess and Angelaki 2003a).

The first few studies that characterized horizontal eye movements typically did so during translation at relatively low frequencies in complete darkness when the eyes adopt a relatively relaxed vergence state (Baloh et al. 1988; Bronstein and Gresty 1988; Buizza et al. 1980; Israël and Berthoz 1989; Melvill Jones et al. 1980; Niven et al. 1966; Skipper and Barnes 1989). As already summarized here, both of these factors have subsequently been shown to have a large influence on the TVOR (Paige 1989; Paige and Tomko 1991a,b; Schwarz and Miles 1991; Schwarz et al. 1989).

Given the relatively recent recognition of the behavioral importance of the TVOR, it is not surprising that the neural correlates for this reflex have remained virtually unexplored. In contrast, the neural substrates of OFR and other short-latency visuo-motor reflexes activated during translation (Miles 1993, 1998) are better understood and have been shown to include the middle superior temporal area of the cortex (MST), the dorsolateral pontine nucleus, and the ventral paraflocculus (Gomi et al. 1998; Kawano and Shidara 1993; Kobayashi et al. 1998; Shidara and Kawano 1993; Takemura et al. 2001a,b). Another reason for the limited understanding of the neural basis for the TVOR is the fact that it is only well developed in primates and humans but remains rudimentary or absent in lateral-eyed species (Baarsma and Collewijn 1975; Dickman and Angelaki 1999; Hess and Dieringer 1990, 1991; Hess et al. 1984; Maruta et al. 2001). In contrast, the highly conserved RVOR has been the subject of many studies in multiple non-primate species. In the following sections, I will summarize some important aspects of the functional neuroanatomy of the otolith system, followed by a brief outline of the current thinking regarding both the necessary sensorimotor transformations and their neural basis in the premotor circuitry of the VORs.

Neuronal architecture of utricular and saccular projections to the oculomotor and spinal motor systems

IPSILATERAL UTRICULO-ABDUCENS PATHWAYS. The "classical" semicircular canal-ocular pathways such as found in the hori-

zonal RVOR are mainly organized within a three-neuron-arc pattern whereby semicircular canal projections through the vestibular nuclei make excitatory connections to contralateral abducens motoneurons and inhibitory connections to ipsilateral abducens neurons (Baker et al. 1969; Precht et al. 1969; Richter and Precht 1968; Schwindt et al. 1973). A very different circuitry has been shown for the otolith-ocular system (Fig. 3). Specifically, abducens motoneurons and internuclear neurons have been shown to receive weak disynaptic (and, in some cases, monosynaptic) EPSPs during ipsilateral utricular nerve stimulation and trisynaptic IPSPs during contralateral utricular nerve stimulation (Imagawa et al. 1995; Schwindt et al. 1973; Uchino et al. 1994, 1996, 1997b). For the utricular projections to the medial rectus, trisynaptic depolarizations have been reported (Fig. 3).

OTOLITH-SPINAL CONNECTIVITY. Whereas direct (i.e., disynaptic) utriculoocular projections are sparse, the opposite is true for the utriculospinal connectivity, where most of the otolith-activated vestibular nuclei neurons appear to participate in vestibulospinal (rather than vestibuloocular) reflexes (Bolton et al. 1992; Ikegami et al. 1994; Kushihiro et al. 1999, 2000; Sato et al. 1996, 1997; Uchino et al. 1997b). Electrical stimulation of the utricular nerve also evokes synaptic potentials in essentially all neck motoneurons (Bolton et al. 1992; Ikegami et al. 1994). The functional significance of these utriculo- and sac-

culo-spinal projections has not been explored. To our knowledge, the only otolith-neck reflex that was ever investigated is a gravity-orientation reflex that was studied in the electromyographic (EMG) activity of neck muscles in decerebrate, canal-plugged animals during vertical plane (i.e., roll) rotations (Schor and Miller 1981; see also Wilson et al. 1995). Bronstein and colleagues have also described human neck EMG activation during free fall (Bisdorff et al. 1999; Ito et al. 1997). Yet whether and how these gravity-sensitive EMG responses function during translation remains unknown. For example, it is possible that a "translational" vestibulocolic reflex exists whereby a horizontal *rotation* of the head could contribute to gaze stabilization during translation. Such a hypothesis is consistent with results during active translation or locomotion in humans, where a head rotation was shown to be a consistent component of gaze stabilization (Bloomberg et al. 1992; Crane and Demer 1997; Demer and Viirre 1996; Medendorp et al. 2002; Moore et al. 1999). Yet when tested during passive translation in monkeys under a limited set of experimental conditions, there was no clear evidence for the existence of such a reflex (Wei and Angelaki 2004). Further experiments are needed to explore this hypothesis and establish a functional role for the strong utriculo-spinal projections. Because our goal is exploring the TVOR, we will not further address these projections here.

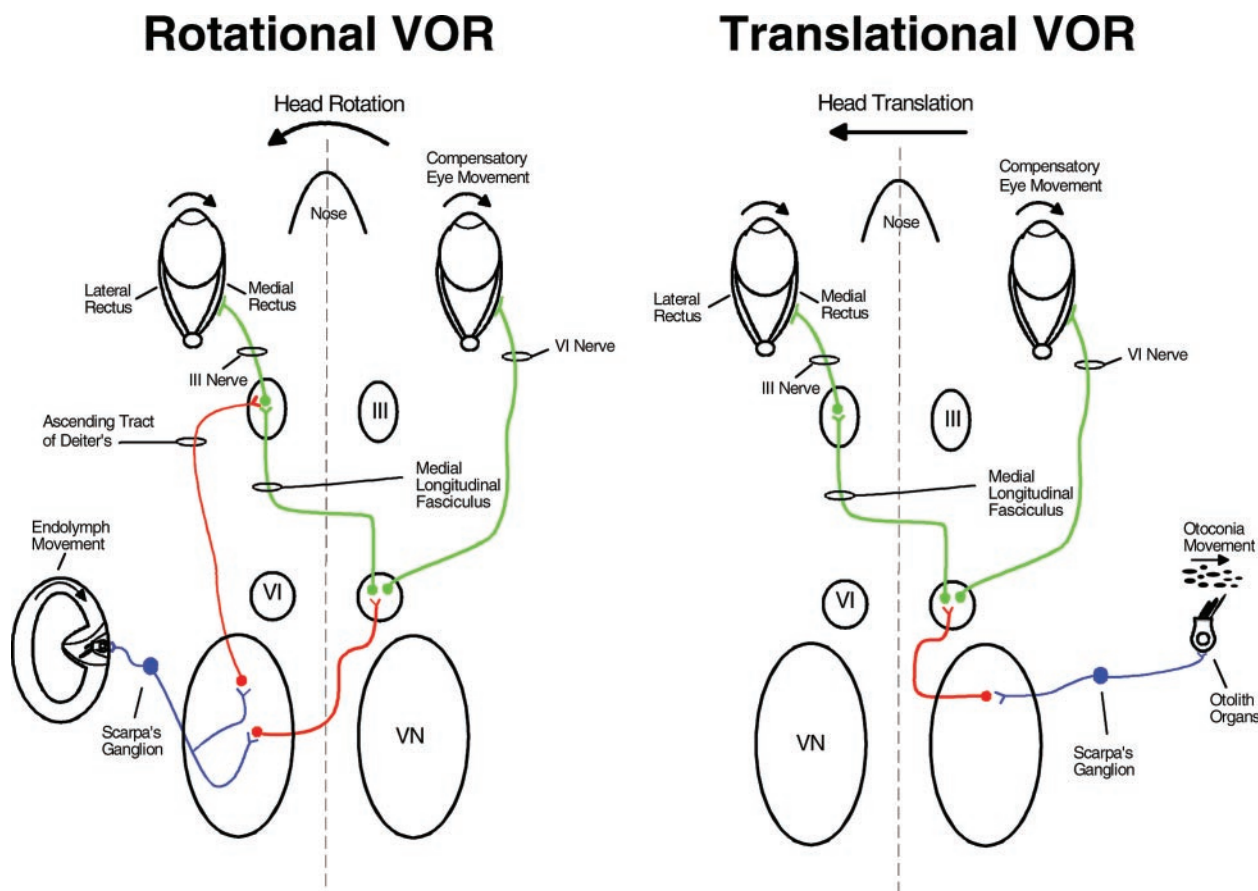


FIG. 3. Schematic illustrating differences in the short-latency utriculo-ocular and semicircular canal-ocular connectivity. Blue lines are used for afferent projections, red lines for 2nd-order connections, and green lines for abducens neuron projections. The shortest-latency projection of the horizontal semicircular canals is to the contralateral abducens and ipsilateral medial rectus nuclei. The shortest-latency excitatory projection of the utricle is to the ipsilateral abducens nucleus (Uchino et al. 1994, 1997b).

Bilateral and bistrilar projections: is there a commissural system for the otolith system?

Earlier work had suggested that otolith-related commissural projections to the lateral vestibular nucleus might have a purely excitatory effect on second-order neurons in the cat (Shimazu and Smith 1971). However, more recent work in the same species has suggested that more than half of the secondary utricular neurons receive commissural inhibition, whereas the remaining do not receive commissural signals at all (Bai et al. 2002; Uchino et al. 2001). On the other hand, almost all the saccular-activated, second-order vestibular neurons exhibit no response to stimulation of the contralateral saccular nerve (Uchino et al. 2001). Because of the opposite polarity of hair cells on the two sides of the striola (Lindeman 1969), a similar effect to commissural inhibition could also take place through cross-striolar inhibition. Indeed, cross-striolar inhibition was observed in two-thirds of saccular-activated (Uchino et al. 1997c) but only one-third of utricular-activated, second-order vestibular nuclei neurons (Ogawa et al. 2000). In contrast, as much as one-third of central utricular-activated cells received the same type of inputs from both sides of the striola, with the remaining one-third of cells receiving inputs from one side only (either medial or lateral). The importance of the presence of excitatory drives from either side of the striola will become clearer in later sections.

Based on the redundant representation of motion direction in the peripheral vestibular organs (Flock 1964; Lindeman 1969) and the presence of cross-striolar convergence, it is possible that a single labyrinth might be sufficient to generate the TVOR. Thus destruction of one labyrinth might not significantly alter the TVOR. Whether this is true or not might depend on the frequency content of the movement. For example, despite a complete loss of the early (<100 ms) transient responses in human subjects with bilateral vestibular lesion and a transient deficit 1 wk after surgery (Lempert et al. 1998), a "relatively normal," bidirectional TVOR was reported in chronic patients with unilateral labyrinthectomy or unilateral vestibular damage (Aw et al. 2003; Bronstein et al. 1991). However, during high-frequency lateral oscillations, a small decrease in TVOR amplitude and its dependence on viewing distance, as well as an eye-velocity misalignment, were observed in rhesus monkeys a few days after unilateral labyrinthectomy (Angelaki et al. 2000c). These deficits gradually recovered over the next months. The decrease in horizontal eye velocity, including a response asymmetry, after unilateral labyrinthectomy was also present during abrupt, high-acceleration transient motion stimuli. During translation toward the lesioned labyrinth, eye velocity was smaller than that during motion towards the intact labyrinth. The closer the target, the larger the decrement in horizontal eye movement. Even though these responses recovered somewhat over time, severe deficits and large asymmetries were still present several months after the operation. Even larger deficits and little response recovery were seen for fore-aft motion stimuli (Angelaki et al. 2000c). The implications of the response asymmetry after unilateral labyrinthectomy regarding the laterality in the organization of utriculo-ocular pathways will be further discussed at the end of this review when we summarize the missing links and outstanding questions.

Related to the question regarding the signals driving the TVOR, there is some evidence that both regular and irregular vestibular afferents might be contributing to the TVOR during high-frequency and -amplitude sinusoidal and transient motion stimuli (Angelaki et al. 2000a). This conclusion was reached by functionally ablating the most irregularly firing afferents using electrical stimulation of the labyrinths bilaterally. These experiments were based on the property that vestibular afferents differ in their sensitivity to labyrinthine electrical stimulation as a function of their regularity of discharge properties (Minor and Goldberg 1991). Anodal (inhibitory) labyrinthine stimulation was shown to decrease TVOR amplitude and its dependence on viewing distance, alter its frequency dependence, as well as result in longer latency and more sluggish responses during abrupt transient movements (Angelaki et al. 2000c). Cathodal (excitatory) currents generally had the opposite effect. In contrast, using lower frequency and lower linear acceleration stimuli during eccentric rotation in squirrel monkeys, Chen-Huang and McCrea (1998a) found no effect of the currents on the TVOR.

Classes of vestibuloocular neurons and comparison of their responses during rotation and translation

Neurons in the vestibular nuclei can be distinguished into four main groups, based on their response properties during steady fixations and saccades, smooth pursuit, and rotational stimulation (Chubb et al. 1984; Cullen and McCrea 1993; Fuchs and Kimm 1975; Keller and Daniels 1975; Keller and Kamath 1975; King et al. 1976; McFarland and Fuchs 1992; Miles 1974; Scudder and Fuchs 1992; Tomlinson and Robinson 1984). For simplicity in the following description, we will focus on the horizontal-eye-movement system. The four cell groups include: 1) position-vestibular-pause (PVP) neurons modulate their firing rates during both pursuit and RVOR suppression, but pause during saccades. Their sensitivities to angular head velocity and to eye velocity are in opposite (complementary) directions such that these signals superimpose during stabilization of an earth-fixed target. PVP cells include units the activities of which modulate either in phase with ipsilateral head velocity during yaw RVOR suppression and contralaterally directed eye velocity during horizontal smooth pursuit (eye-contra PVP) or in phase with contralateral head velocity during RVOR suppression and ipsilaterally directed eye velocity during smooth pursuit (eye-ipsi PVP). (Please note that the eye-contra/eye-ipsi description used here corresponds to the traditional type I or II categorization used by previous studies.) 2) Eye-head (EH) neurons also modulate their firing rates during both pursuit and RVOR suppression. However, in contrast to PVP neurons, their sensitivity to head velocity during RVOR suppression and to eye velocity during smooth pursuit are in the same direction, such that the two signals oppose each other during rotation while stabilizing an earth-fixed target. This group includes cells with ipsilaterally directed eye- and head-velocity sensitivities (eye-ipsi EH) as well as cells with contralaterally directed eye- and head-velocity sensitivities (eye-contra EH). 3) Burst-tonic (BT) neurons do not respond during yaw RVOR suppression but modulate during steady fixations, saccades, and smooth-pursuit eye movements. 4) Vestibular-only (VO) neurons include all cells

that do not exhibit any eye-movement sensitivity but modulate during either rotational or translational head movements.

Among the four classes, BT, PVP, and EH cells have been shown to provide the primary premotor drive for slow eye movements by making direct projections to either the contralateral or ipsilateral abducens nucleus (McCrea et al. 1987; Scudder and Fuchs 1992) or to the ipsilateral oculomotor nucleus via the ascending tract of Dieters (Chen-Huang and McCrea 1998b). Of these three groups, it is only the BT neurons that might be considered to carry a fully transformed “motor” signal for the RVOR in the sense that they carry signals that are strongly correlated with the associated eye movement during all behavioral conditions. Specifically, BT neurons carry signals related to deviations in ocular position and modulate similarly during all slow eye movements. They do not, however, change their firing rates during head rotation when the eyes remain still. In contrast, PVP and EH neurons discharge differently during smooth pursuit and head rotations even with identical ocular deviations. Moreover, they exhibit clear response modulations during RVOR suppression in the absence of compensatory eye movements. In this sense, although many are premotor cells, both PVP and EH neurons can be considered to carry information that is intermediate in the transformation of sensory signals (semicircular canal) to motor output during rotation. As will be summarized in the following paragraphs, this is not the case for the TVOR.

In addition to these four classes of cells, floccular-target-neurons (FTNs) in alert primates have been identified as a subgroup of VOR interneurons that receive direct inhibition from Purkinje-cells in the flocculus/ventral paraflocculus (FL/VPF) and participate in motor learning (Lisberger and Pavelko 1988; Lisberger et al. 1994; Partsalis et al. 1995a,b; Zhang et al. 1995a,b). The properties of FTNs resemble those of EH cells, although the extent of overlap in these two populations remains relatively unexplored. For the horizontal-eye-movement system, both cat and rabbit FTNs project to the ipsilateral and not to the contralateral abducens nucleus (Highstein 1973; Ito et al. 1977; Sato et al. 1988). Even though FTN response behavior during saccades and VOR suppression could vary greatly, their unique characteristics include a strong eye-velocity sensitivity with a small and often non-linear eye-position sensitivity. FPNs, on the other hand, have been described as cells with small eye-movement sensitivity and represent distinctly different and nonoverlapping populations from FTNs (Nagao et al. 1997; Zhang et al. 1993, 1995a,b). Neuroanatomical and electrophysiological studies have identified FTNs in the rostral medial and ventrolateral vestibular nuclei (Langer et al. 1985a; Lisberger et al. 1994; Nagao et al. 1997; Sato et al. 1988). FPNs, on the other hand, are located in mostly nonoverlapping areas, including the caudal medial and inferior vestibular nuclei (Langer et al. 1985b).

All three neuron groups with eye-movement-related activities known to participate in the generation of the RVOR (PVP, EH, and BT cells) also participate in the generation of the TVOR (Angelaki et al. 2001; Chen-Huang and McCrea 1999b; King et al. 2003; McConville et al. 1996). Yet there is growing evidence that the vestibular signal flow into the premotor VOR network might be different for the RVOR and the TVOR. Specifically, the firing rates of some cells that carry both sensory head movement and motor-like signals during rotation (PVP and EH cells) are more strongly related to the motor

output than to the vestibular sensory signal during translation. That is, the main secondary neuron in the disynaptic RVOR pathways (eye-*contra* PVP cell) that exhibits a robust modulation during RVOR suppression does not modulate during TVOR suppression. In contrast, the majority of eye-*ipsi* cells (mostly eye-*ipsi* PVP but also a few eye-*ipsi* EH cells) exhibit a clear response modulation during TVOR suppression in the absence of eye movements (Angelaki et al. 2001; King et al. 2003; Meng and Angelaki 2003). Further differences in the neural response dynamics prompted Angelaki et al. (2001) to propose that otolith afferents might directly project onto eye-*ipsi* but not eye-*contra* cells (Fig. 4). Such a hypothesis is totally compatible with the known differences in the neuroanatomy of otolith-ocular vs. canal-ocular pathways, including the disynaptic ipsilateral utriculo-abducens connectivity as eye-*ipsi* cells could have weak excitatory projections to the ipsilateral abducens. In contrast, eye-*contra* PVP cells, which are the secondary interneuron in the semicircular canal-ocular pathways, most likely do *not* receive direct primary otolith afferent input because their direct excitatory projections to the contralateral abducens nucleus (McCrea et al. 1987; Scudder and Fuchs 1992) would be incompatible with the utriculo-abducens connectivity (Uchino et al. 1994, 1997b).

It is important to point out that even if the shortest-latency utriculo-ocular pathway is ipsilateral and does not involve the eye-*contra* PVP neurons, this does not imply that the eye-*contra* PVP pathway plays an insignificant role in the TVOR. In fact, eye-*contra* PVP neurons represent the main oculomotor projections of the vestibular nuclei (McCrea et al. 1987; Scudder and Fuchs 1992). Therefore eye-*contra* PVP cells could still carry the bulk of the motor drive for the generation of the TVOR (Angelaki et al. 2001; Chen-Huang and McCrea 1999b). This role could also be served by other contralaterally projecting premotor neurons, e.g., the BT cells of the vestibular and prepositus nuclei, but is unlikely due to the weak ipsilateral utriculo-abducens projections. We will return to this issue at the end of the review when we attempt to summarize the puzzles of what we know and what we do not understand about this oculomotor system.

In summary, contrary to the RVOR where the shortest-latency, three-neuron-arc pathway seems to be the most abundant and strongest projection, the bulk of the signals to move

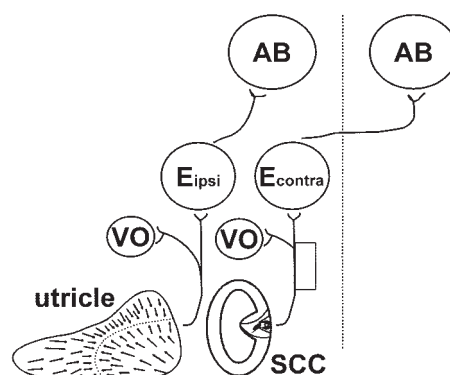


FIG. 4. Sensory otolith signals join the premotor VOR circuitry through projections to vestibular-only (VO) and eye-*ipsi* (E_{ipsi}) cells. In contrast, no direct utricular afferent projections seem to exist to eye-*contra* (E_{contra}) cells, which receive direct monosynaptic projections only from the semicircular canals (SCC). AB, abducens nucleus. The vertical dashed line represents the midline. Polysynaptic projections are not illustrated.

the eyes during translation may utilize polysynaptic pathways. The existence of weak short-latency ipsilateral and strong, longer-latency contralateral pathways could still elicit eye movements at a short latency, ~ 10 – 12 ms, as has been shown for the TVOR (Angelaki and McHenry 1999). Why would the TVOR utilize polysynaptic pathways to the motoneurons? Why don't primary otolith afferents synapse directly onto eye-contra PVP neurons? As will be summarized in the following text, the answer to this question lies in the extensive computational steps that are necessary to convert sensory vestibular signals into eye-movement commands during translation. For simplicity, I will continue to focus on the horizontal system and more specifically the utriculo-abducens pathways.

Is there a role for the cerebellar flocculus/ventral paraflocculus in the TVOR?

Before discussing the extensive processing that needs to take place, I will briefly summarize the evidence supporting a potential role of the cerebellar FL/VPFL. First and foremost, the FL/VPF and presumably also the FL/VPF-receiving areas of the vestibular nuclei are involved in the processing of OFR which is postulated to work closely in synergy with the TVOR, including sharing common processing stages such as their dependence on viewing distance (Miles 1993, 1998; Schwarz et al. 1989). Second, FL/VPF Purkinje cells modulate their firing rates at short latencies during translation (Snyder and King 1996). Third, deficits in the TVOR, particularly decreases in its dependence on viewing distance, have been reported in patients with cerebellar deficits (Baloh et al. 1995; Crane et al. 2000). A direct, histologically verified FL/VPFL lesion in monkeys to specifically investigate the importance of the FL/VPF in the properties of the reflex has not been yet performed. Preliminary data from David Zee's laboratory have reported that surgical lesions of the FL/VPF result in strongly compromised TVOR responses in rhesus monkeys (Zee et al. 2002).

The otolith signals to the flocculus could either be direct or indirect through the vestibular nuclei. Afferent projections to the FL/VPF are numerous from the vestibular nuclei, particularly from the caudal medial and inferior subdivisions (Langer et al. 1985b). These FL/VPFL-projecting areas considerably overlap with those that receive direct utricular afferent input (Imagawa et al. 1995; Siegborn and Grant 1983). However, Uchino and colleagues did not find second-order utricular cells that were floccular projecting neurons in the cat vestibular nuclei (although the ventral paraflocculus was not tested) (Y. Uchino, personal communication). There exists recent evidence suggesting the existence of weak utricular afferent projections to the FL/VPFL in the macaque (Newlands et al. 2003; but see Langer et al. 1985b; Nagao et al. 1997). Interestingly, these utricular projections in the primate FL/VPFL were not present in the gerbil (Newlands et al. 2002; Purcell and Perachio 2001), perhaps reflecting the evolutionary change in otolith-ocular function.

It is believed that at least some FTNs are premotor cells that project directly to motoneurons. The most direct evidence comes from work in non-primate species demonstrating that horizontal FTNs project to the ipsilateral and not to the contralateral abducens nucleus (Baker et al. 1972; Highstein 1973; Ito et al. 1977; Sato et al. 1988). In primates, some eye-contra EH cells have also been shown to make inhibitory connections

with the ipsilateral abducens nucleus (Scudder and Fuchs 1992). In contrast, the output projections of eye-ipsi EH neurons (which could also be FTNs) are unknown. A previous study failed to find connections between the eye-ipsi EH cells and either the abducens nuclei or the vestibular nerves (Scudder and Fuchs 1992). McCrea et al. (1987) did stain two ipsilateral eye-movement cells that exhibited bursts in activity during saccades and at least one received monosynaptic excitation from the ipsilateral labyrinth. These neurons were not tested during rotation, but they could have been eye-ipsi EH cells and/or eye-ipsi FTNs. Based on their eye movement sensitivity, if they are premotor neurons, eye-ipsi EH or FTN cells should be excitatory to the ipsilateral abducens nucleus.

The secondary neurons in the ipsilateral utriculo-ocular excitatory pathway have been identified in the ventrolateral vestibular nuclei (Uchino et al. 1994), an area that overlaps with the location of horizontal FTNs (Langer et al. 1985a; Lisberger et al. 1994; Nagao et al. 1997; Sato et al. 1988). Thus it is possible that at least some of the second-order utricular-activated vestibular nuclei neurons might be FTNs. Although totally speculative at this point, this possibility is worth considering as we try to unmask the challenges and complexities of the TVOR.

Sensorimotor transformations

Figure 5 summarizes the two sets of computational steps needed to convert sensory vestibular signals into eye-movement commands for the TVOR. The first step involves an estimation of the actual translation (more on this will follow in the next section). The second step includes three basic transformations of sensory information encoding inertial (translational) motion to ensure appropriate motor commands. These include: the dynamic transformation of linear acceleration oto-

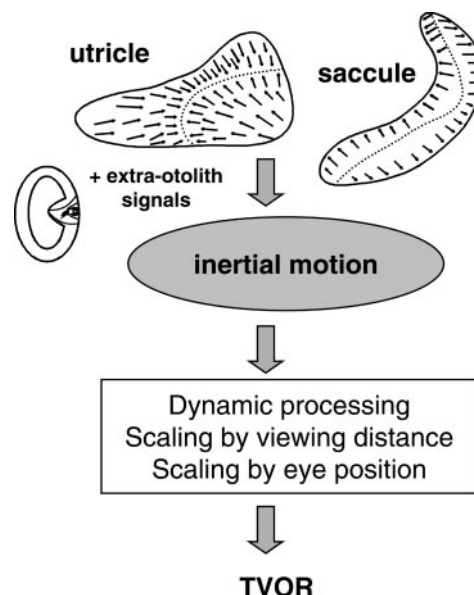


FIG. 5. Schematic illustrating the main sensorimotor transformations in the TVOR. Otolith signals must first be combined with extra-otolith information (e.g., arising from the semicircular canals) to compute a central estimate of inertial (translational) motion. Subsequently, these signals need to be further processed dynamically to generate temporally appropriate motoneuron commands as well as be scaled by a neural estimate of viewing distance and eye position to implement Eq. 1.

lith signals into eye-position commands; the modulation of sensory signals in the TVOR pathways as a function of target distance; and the transformation of head-fixed vestibular signals into the foveocentric organization of the TVOR. Here we expand on each of these, with a summary of what is currently known and what experiments have yet to be done.

TEMPORAL TRANSFORMATION: IS THERE A MISSING, ADDITIONAL INTEGRATOR IN THE TVOR? The signals carried by primary semicircular canal and otolith afferents differ in their dynamic properties. Specifically, primary otolith afferents encode linear acceleration, whereas semicircular canal afferents code most closely for head velocity in a broad frequency range (Fernandez and Goldberg 1971, 1976c). Thus a requirement for additional low-pass filtering of otolith as compared to canal signals in the VOR pathways has been suggested. Traditionally, it has been thought that a separate, additional integration or low-pass filtering of otolith signals exists prior to the convergence on a neural integrator and the central network shared with the RVOR (Paige and Tomko 1991b; Telford et al. 1998). An alternative hypothesis has been recently introduced that suggests no additional central filtering of otolith signals in the TVOR. Rather, otolith and canal afferents were proposed to converge onto a shared but distributed velocity-to-position neural integrator network at distinct sites such that sensory information from the semicircular canals is only integrated once centrally while simultaneously otolith signal information is integrated twice (the second integration provided by the eye plant itself) (Green and Galiana 1998). This proposal is further developed in the next section, which addresses the proposed dynamic models for the TVOR. However, it is important to mention that this issue is still open to debate and further experimentation. Yet at least one aspect of this hypothesis seems to be true: a differential projection of otolith and canal signals onto central eye-ipsi and eye-contra cells is supported by both anatomical and physiological observations (Angelaki et al. 2001).

CENTRAL CORRELATES OF VIEWING DISTANCE DEPENDENCE. The first studies examining how vestibular nuclei neurons vary their firing rates as a function of viewing distance utilized either relatively low-frequency eccentric rotation (Chen-Huang and McCrea 1998b, 1999a,b; McConville et al. 1996) or low-frequency translation (Angelaki and Dickman 2002) in the presence of visual feedback. These studies reported that all three eye-movement-sensitive cell classes (PVP, EH, and BT cells) change their firing rates as a function of viewing distance. Those whose firing rates changed the most with viewing distance during centered and eccentric rotations were the EH cells (Chen-Huang and McCrea 1999b; McConville et al. 1996). More recently, vestibular nuclei responses have also been tested during high-frequency translation in darkness; it was primarily EH and BT cells that exhibited the largest changes with vergence angle (King et al. 2003; Meng and Angelaki 2003). Despite these few reports, the site(s) and dynamics of TVOR gain modulation by target distance remain unknown. Given the extensive interconnections between VOR neurons, it is unclear if the observed sinusoidal responses represent feed-forward "motor commands" as opposed to efference feedback. It is important that future studies compare the latency of the changes in neural firing rates to causally relate observed neural firing rate changes to those that can

"drive" the observed eye movements (King et al. 2003). Furthermore, because it has been shown that the scaling by viewing distance is shared by both the TVOR and the OFR (Schwarz et al. 1989), but not smooth-pursuit eye movements (Zhou et al. 2002), it is also important to test the viewing distance dependence of neurons during *both* TVOR and OFR. If it is indeed true that both reflexes share a common viewing distance-dependent processing and because the pathway eliciting OFR has been shown to involve the FL/VPF (Gomi et al. 1998; Kawano and Shidara 1993), FTNs should be the cell population that exhibits large and short-latency viewing distance-dependent changes (as has actually been reported for EH cells during steady-state sinusoidal motion) (Chen-Huang and McCrea 1999a,b; King et al. 2003; McConville et al. 1996; Meng and Angelaki 2003). If so, does the scaling by viewing distance take place in the FL/VPF itself (e.g., on Purkinje cells) or at the level of FTNs? A brief report by Snyder and King (1996) suggests that the shortest-latency ensemble response of a Purkinje cell population did not scale by viewing distance during eccentric rotation even though the sensitivity to translation signals was unquestionable. Although preliminary, this report supports the hypothesis that the actual scaling by viewing distance takes place in the brain stem.

TRANSFORMATION OF HEAD-FIXED VESTIBULAR SIGNALS INTO THE FOVEOCENTRIC ORGANIZATION OF THE TVOR. Where and how the eye-position dependence in Eq. 1 is implemented in the premotor circuitry remains unknown. Other than some preliminary results (Angelaki and Dickman 2002), studies have yet to be performed to address this issue. These experiments must vary target eccentricity and test for neural response changes as a function of eye position. It is optimal if this is done during fore-aft motion, as it is the TVOR during translation along axes closely parallel to gaze that exhibits the largest dependence on eye position (Eq. 1, see also Fig. 2). For neural signals to be appropriate to drive the oculomotor system and generate the gaze dependence of the TVOR, peak neural firing rate and phase should be a function of eye position, similarly as eye movements, during translation. As previously reported, horizontal eye velocity during fore-aft motion is zero during straight-ahead gaze, whereas its dependence on eye position is manifested as a "V-shaped" curve for the response amplitude and a 180°-shift in phase for rightward and leftward eye positions (Fig. 6A, ○; see also Fig. 2). The majority of eye-contra and a few eye-ipsi cells tested during fore-aft motion for different eye positions changed their firing rate modulation phase for targets to the left and to the right and exhibited an eye-position dependence that paralleled that of horizontal eye velocity (Angelaki and Dickman 2002; Meng and Angelaki 2003). The dependence of neural firing rates on eye position extended to different heading directions (Fig. 6A). For heading directions 10° to the left and to the right of the fore-aft axis, the V-shaped curve and the phase reversal shifted accordingly along the eye-position axis. When replotted relative to the difference angle ($\alpha - \theta$), the three heading direction data sets were superimposed (Fig. 6B), suggesting that the cell modulated according to Eq. 1.

Similar to the dependence on vergence angle, future work must examine the timing of the neural response dependence on

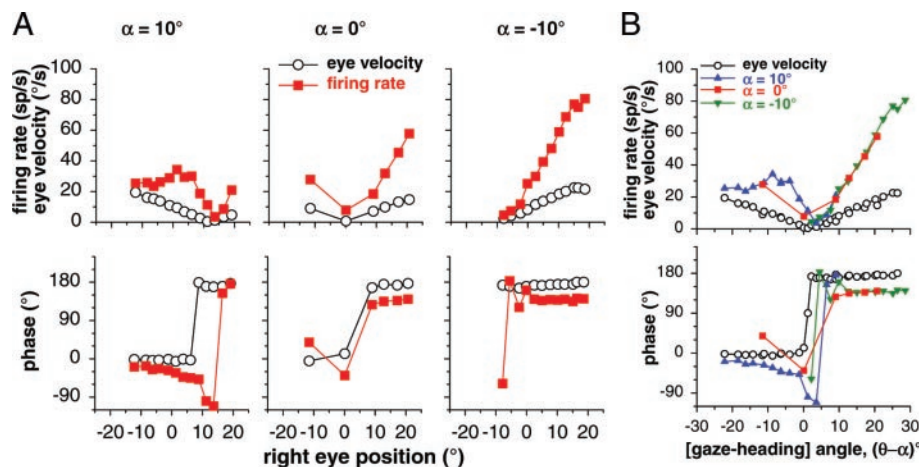


FIG. 6. Dependence of the firing rate of an eye-in-head (EH) cell as a function of eye position and heading direction (0.5 Hz). A: eye velocity ($\circ$) and peak neural firing rate ($\blacksquare$) are plotted vs. eye position for 3 different heading directions; $\alpha = 0^\circ$ (along the animal's fore-aft arc; middle), $\alpha = -10^\circ$ (along an axis rotated 10° clockwise; left), and $\alpha = 10^\circ$ (along an axis rotated 10° counter-clockwise; right). B: the same data have been re-plotted, now relative to the difference angle between gaze and heading directions ($\theta - \alpha$). The data for the different heading directions superimpose, suggesting that neural firing rates encode the respective dependence of eye velocity (see Eq. 1). Modified from Angelaki and Dickman (2002).

eye position using transient motion stimuli. In contrast to the viewing distance-dependent modulation that we hypothesize takes place in the brain stem *downstream* of the site of convergence between TVOR and OFR signals, the eye-position-dependent transformation might occur *upstream* of their site of converge. FL/VPFL neuron activities have not yet been examined during fore-aft motion as a function of eye position.

Computational models of the TVOR

There exist a few models that deal primarily with only one aspect of the sensorimotor transformations needed to generate an eye movement from primary otolith afferent signals: the temporal processing to convert linear acceleration into eye-position commands. The basic idea behind these models, which were independently proposed by two different groups (Green and Galiana 1998; Musallam and Tomlinson 1999), is the following: the RVOR and TVOR share the neural integrator pathway, yet they have completely separate "direct" pathways.

The direct pathway for the RVOR is strong, whereas the direct pathway for the TVOR is very weak (Fig. 7A, blue and red projections, respectively). As a result, the strong direct RVOR projection generates a lead that can compensate for the low-pass dynamics of the eye plant (Robinson 1981; Skavenski and Robinson 1973). In contrast, the direct TVOR pathway is too weak, thus the low-pass filter properties of the eye plant, which remain uncompensated, can provide the second integration for the TVOR. This second integration can only be effective at high frequencies, above the dominant time constant of the eye plant (Green and Galiana 1998), a fact that is consistent with the dynamics of the TVOR (Angelaki 1998; Paige and Tomko 1991a; Telford et al. 1997). This proposal is compatible with the existing neuroanatomy if the direct excitatory canal pathway is contralateral and the direct excitatory otolith pathway is ipsilaterally projecting (e.g., Uchino et al. 1997a). This conceptual organization of the RVOR and TVOR is also consistent with the observed differences in translational response proper-

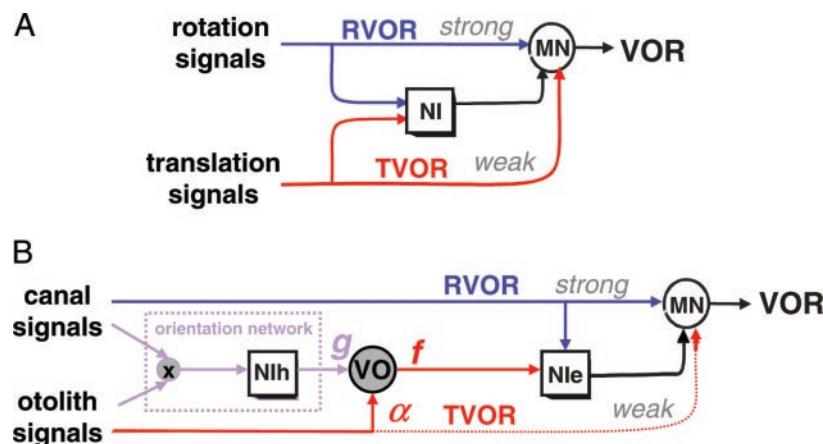


FIG. 7. Proposed flow diagrams for the VORs. A: a schematic showing the basic idea behind the proposed temporal processing in the RVOR and the TVOR. Briefly, both rotation and translation signals are processed by a common neural integrator (NI) (Robinson 1981; Skavenski and Robinson 1973), but the direct projection of translation signals to the motoneurons (MN) is very weak as compared to a strong direct projection of rotation signals (Angelaki et al. 2001; Green and Galiana 1998; Musallam and Tomlinson 1999). B: a similar diagram now expanded to include the computation of inertial (translation) motion acceleration, f , through a multiplicative interaction of semicircular canal and otolith afferent information. This additional processing (*orientation network* in purple) computes an internal estimate of gravity, g , in head coordinates. This processing includes a head-velocity-to-position neural integrator (NIh) that integrates angular head velocity. This integrator is distinct from the eye-velocity-to-position integrator (Nle) that is shared with the RVOR. Vestibular-only (VO) cells subsequently subtract this signal from the net gravito-inertial acceleration, α , to extract an internal estimate of translation, f (Green and Angelaki 2003; 2004; Merfeld and Zupan 2002).

ties in eye-contra and eye-ipsi cells (Fig. 4). Angelaki et al. (2001) have described a feedback network, expanded from the one originally proposed by Green and Galiana (1998), to include these differential projections in a network that includes both eye-ipsi and eye-contra cells.

Sensory ambiguity for detecting translation

In the discussion so far, it was implied that the translational signals needed for the generation of the TVOR arise exclusively from the otolith organs. However, as will be summarized here, this would be physically impossible because otolith afferents respond similarly to inertial and gravitational accelerations (Einstein's equivalence principle; Einstein 1908). Thus the otolith system provides inherently ambiguous sensory information, as the encoded acceleration could have been generated during either actual translation or a head reorientation relative to gravity (Angelaki and Dickman 2000; Fernandez and Goldberg 1976a,b). How does the brain deal with this ambiguity? One solution would be to either ignore the problem all together or use conceptually easy but mathematically inappropriate approximations and, as a result, compromise motor and/or perceptual performance. The second solution would be to attempt to resolve the ambiguity by simultaneously utilizing extra-otolith cues that might be available. As summarized here, there is growing evidence that, whenever possible, the brain attempts to utilize extra-otolith sensory information to extract a central estimate of translational motion.

The question of whether and how translations and tilts are distinguished by the brain has been a source of continued debate. For example, it has been suggested that internal estimates of translational motion and head tilt need not be explicitly constructed, but that an approximation is made where low-frequency linear accelerations could be interpreted as tilts and high-frequency stimuli as translations (Paige and Tomko 1991a; Telford et al. 1997). This proposal has been popular in the vestibular community as it eliminates the need for complex vector differential equations that are required to actually solve the ambiguity problem mathematically. The fact that low-frequency linear accelerations are typically perceived as tilts even when the head is not reoriented relative to gravity (Clark and Graybiel, 1966; Merfeld et al. 2003; Seidman et al. 1998) has been used as support for this hypothesis. This issue remains a controversial topic in vestibular research.

A key problem with the "frequency-segregation" hypothesis is that it does not explain how mid-high-frequency translations are distinguished from tilts. Because in this frequency range angular motion cues from the canals can provide complementary sensory information, why shouldn't a smart brain maximize its sensory processing to improve performance rather than rely on suboptimal approximations? According to the simplistic approximation, high-frequency linear accelerations would always be interpreted as translation. Thus during high-frequency rotations that change the orientation of our head relative to gravity, we would perceive that we are being simultaneously rolled and translated. Obviously, this is not what is perceived during these very natural head movements. In addition, according to the frequency-segregation hypothesis, our eyes would simultaneously rotate torsionally (RVOR) and horizontally (TVOR) during roll tilt, the latter component being particularly large for near targets. Several studies have shown

that this does not occur and that the primate VOR distinguishes between tilts and translation (Angelaki et al. 1999; Green and Angelaki 2003; Hess and Angelaki 1999), although this issue remains unresolved for the human VOR (Park et al. 2003; but see also Merfeld et al. 2001; Zupan et al. 2000).

The idea that the brain integrates information from both otolith and semicircular canal afferents to differentiate translation from tilt has existed for a long time (Guedry 1974; Mayne 1974; Young 1984). Recently, it has been further supported by several investigations (Angelaki et al. 1999; Glasauer and Merfeld 1997; Green and Angelaki 2003, 2004; Merfeld 1995; Merfeld and Zupan 2002; Merfeld et al. 1999; Mergner and Glasauer 1999; Zupan et al. 2002). Theoretical studies have proposed that the brain can explicitly construct internal estimates of gravity and translational acceleration by effectively solving a vector differential equation that relies on an estimate of the rotational velocity of the head (Merfeld 1995; Merfeld and Zupan 2002; Mergner and Glasauer 1999; Zupan et al. 2002). Solving this equation to estimate the instantaneous gravity vector in head coordinates implies a temporal integration of angular velocity, suggesting the existence of a head rotation integrator in the VOR pathway (Fig. 7B, Nih) (Green and Angelaki 2003, 2004). This gravity estimate can then be combined with the net gravito-inertial acceleration sensed by the otoliths to extract the translational component of the movement (Fig. 7B, purple pathway).

According to this model (summarized in the schematic of Fig. 7B), it would be predicted that the canals should then also contribute to driving the TVOR (Glasauer and Merfeld 1997; Merfeld 1995; Merfeld and Zupan 2002; Mergner and Glasauer 1999; Zupan et al. 2002). This has been confirmed experimentally using combinations of roll tilt and translation to examine oculomotor responses in the absence of a net lateral acceleration stimulus to the otoliths. Horizontal ocular responses appropriately directed to compensate for the translational component of motion were elicited even in the absence of a dynamically relevant otolith sensory stimulus (Angelaki et al. 1999; Green and Angelaki 2003). The necessary contribution of semicircular canal cues to these horizontal eye movements was directly demonstrated by the fact that they were no longer evoked in canal-plugged animals (Angelaki et al. 1999). Recently, it has been shown that these canal-driven responses represent an extra-otolith TVOR that exhibits dynamic properties and a dependency on viewing distance similar to those of the purely otolith-driven reflex (Green and Angelaki 2003). Quantitative analyses further showed that this extra-otolith driven TVOR was best correlated with angular head *position*, thus verifying that angular velocity signals from the semicircular canals are processed by an additional neural integrator in the TVOR pathways (Fig. 7B) (Green and Angelaki 2003).

How single neurons implement these computations was explored in a recent theoretical study that proposed how a physiologically relevant, nonlinear integrative neural network could be used to perform the required computations for translational motion detection (Green and Angelaki 2004). In this network, the vector differential equations were implemented as a set of multiplicative interactions where semicircular canal signals were multiplied by a net acceleration signal before being fed into an integrative network that computed a central estimate of gravity (Fig. 7B, purple pathway). The predictions of this model are consistent with, and in fact provide an explanation

for, the large diversity in neural response properties during translation and the seemingly complex patterns of otolith/canal convergence that occurs centrally (Angelaki and Dickman 2000; Dickman and Angelaki 2002; Zhou et al. 2001). Furthermore, model simulations also predict the existence of tilt neurons with low-pass filter properties to translation as indeed reported experimentally (Zhou et al. 2000). The computation of translational movement information seems to occur in VO cell populations in the vestibular nuclei, the rostral fastigial nuclei, and the cerebellar nodulus. Neurons in these areas have been shown to appropriately combine temporally processed semi-circular canal and otolith signals as required for an internal model calculation of translational motion (Angelaki et al. 2003).

Summary of premotor cells' role in the TVOR

Although fundamental questions remain unanswered, in the following paragraphs, I will attempt to put together a brief summary of the underlying processing, even if the overall picture is not yet complete. As has been the case throughout this review, I continue to focus on the horizontal system and more specifically the utriculo-abducens pathways.

EYE-CONTRA PVP NEURONS. Experimental evidence to date suggests that these cells, which represent the main interneuron in the three-neuron arc RVOR pathway, do not serve a similar role for the TVOR. Specifically, it appears unlikely that eye-contra PVP neurons receive direct otolith afferent inputs. In addition, at least under steady-state conditions, the firing rates of these cells reflect "motor" signals that are more closely related to the elicited eye movement than the sensory vestibular stimulus. Accordingly, they exhibit the appropriate dependence on eye position and viewing distance, as outlined by Eq. 1 and they do not typically modulate during TVOR suppression in the absence of eye movements.

EYE-IPSI PVP NEURONS. It is possible that these cells represent the main interneuron in the weak three-neuron arc of the TVOR (Uchino et al. 1997a). In sharp contrast to eye-contra PVP cells, the modulation of eye-ipsi PVP neurons does not reflect the motor signals needed for the generation of the TVOR. Accordingly, the firing rates of most cells modulate during far target viewing and do not exhibit a strong scaling with either vergence angle or eye position. The lack of tuning according to Eq. 1 as well as the fact that most eye-ipsi PVP cells exhibit a strong modulation during TVOR suppression in the absence of eye movements suggests a sensory, rather than motor, function of these cells.

EYE-CONTRA EH NEURONS. Eye-contra EH neurons behave during translation similar to eye-contra PVP cells, having properties closer to motor than sensory. Accordingly, they exhibit the appropriate dependence on eye position and viewing distance as outlined by Eq. 1 and the majority do not modulate during TVOR suppression in the absence of eye movements.

EYE-IPSI EH NEURONS. Among all premotor neuron classes, those with the most diverse behaviour during translation are the eye-ipsi EH cells. The possibility that at least some of the eye-ipsi EH neurons receive direct primary otolith input can not be excluded. The response reversal during stable gaze/suppression conditions strongly argues for a second input from

the FL/VPFL. It also remains possible that some eye-ipsi EH cells might be both secondary utriculo-ocular neurons and FTNs.

BURST-TONIC NEURONS. Responses of BT neurons are similar to those of eye-contra PVP cells during translation. Accordingly, most BT neurons do not modulate during TVOR suppression and their properties as a function of viewing distance and eye position routinely paralleled those of eye movements. From this, we would also propose that, similarly to eye-contra PVP neurons, BT cells do not receive a direct otolith afferent input. However, Chen-Huang and McCrea (1999b) reported that there is a significant viewing-distance-sensitive otolith component on BT cells that could not be simply attributed to eye-movement signals. In line with this motion, some eye contra BT neurons could be monosynaptically activated after electrical stimulation of the ipsilateral vestibular nerve (Chen-Huang and McCrea 1999a; Scudder and Fuchs 1992). Yet, a sensory utricular afferent signal has yet to be reported in BT cell responses during translation.

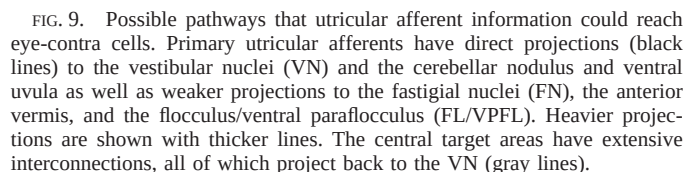
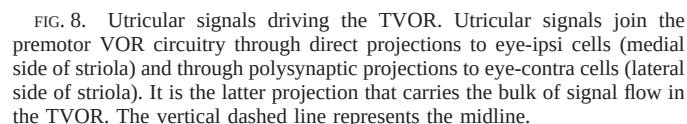
Missing links

Given the multiple computational steps needed, it is now better appreciated why polysynaptic pathways probably represent the main signal flow for the TVOR. Both the calculation of translation motion itself, not to mention the neural implementation of the eye position and vergence angle dependence of Eq. 1, point to the realization that the disynaptic utriculo-abducens pathway, even if it were strong, would fall short of generating an appropriate TVOR. Then the natural question to ask is why is there a weak disynaptic pathway at all? Why aren't all utriculo-abducens projections polysynaptic? Insight to this question comes from the modeling studies. A weak direct projection of utricular signals to the abducens could shape the dynamics of the TVOR (Angelaki et al. 2001; Green and Galiana 1998; Musallam and Tomlinson 1999).

Another related issue that poses great challenges and deserves further consideration is the laterality of the utricular signals that generate the TVOR. The excitatory short-latency utricular input to the abducens probably arises from the medial side of the striola (Ogawa et al. 2000) to have the correct directional sensitivity to contribute to a compensatory eye movement (Fig. 8). For example, signals originating from the medial side of the striola (increasing their firing rates for contralateral translation) make excitatory connections with the ipsilateral abducens, thus contributing to the generation of ipsilaterally directed eye movements. If this was the major signal flow in the reflex, unilateral labyrinthectomy would then selectively compromise the TVOR during motion away from the lesion side. However, the opposite result has been reported in both humans and monkeys: after unilateral labyrinthectomy, it is motion toward, not away, from the lesion side that is most severely compromised (Angelaki et al. 2000c; Lempert et al. 1998). Therefore lesion results would argue that it is the *lateral side* of the striola (i.e., the side excited during ipsilateral movements) that generates the major drive to the TVOR (Fig. 8).

If the weak disynaptic connections described by Uchino and colleagues (Uchino et al. 1997a) and the corresponding sensory-like physiological properties of the corresponding eye-ipsi

The potential role of the cerebellar FL/VPFL also remains a puzzle. As mentioned in the preceding paragraphs, Purkinje cells in these areas have been reported to change their firing rates at short latency in response to translation (Snyder and King 1996) and patients with FL/VPFL lesions were reported to have large deficits in the TVOR (Baloh et al. 1995; Crane et al. 2000). Yet, it is presently unclear how utricular and/or saccular signals reach the FL/VPFL. A utricular afferent projection that has been reported in monkeys (Newlands et al. 2003) is rather weak and does not appear to be true for the sacculus. Second-order utricular or saccular neurons in the vestibular nuclei also do not seem to provide a large projection system to the FL/VPFL (Y. Uchino, personal communication). Polysynaptic otolith signals could reach the FL/VPFL through the vestibular nuclei, although their influence on the early TVOR responses might be questionable.



We have come long ways during the past decades in unmasking the mysteries of the functional goals of the main vestibulo-motor system for foveal vision and stereovision. Signals from both the otolith and the semicircular canals are important for the generation of the TVOR. Neuroanatomical and neurophysiological data have provided strong evidence that translational but not rotational motion information has been dynamically transformed into motor-like signals at the level of eye-contralateral PVP neurons and that these neurons receive direct canal but not otolith afferent signals. In contrast, the responses of eye-ipsilateral PVP and EH cells seem to represent an intermediate signal that is neither purely motor nor sensory, suggesting that these cell types may be functionally located within, rather than at the motor end, of the underlying sensorimotor transformations for the TVOR. Growing evidence based on lesion and neuroanatomical studies suggest that it is polysynaptic utricular projections from the lateral side of the striola to the contralateral abducens nucleus that mediate the TVOR. Yet more remains to be discovered about its neural organization. At present, several pieces of the puzzle remain missing and await future explorations.

I thank A. Green, A. Haque, and M. King for commenting on the manuscript.

This work was supported by National Eye Institute Grant EY-12814.

Angelaki DE. Three-dimensional organization of otolith-ocular reflexes in rhesus monkeys. III. Responses to translation. *J Neurophysiol* 80: 680–695, 1998.

Angelaki DE. Dynamics and viewing distance dependence of eye movements during transient lateral motion. *Arch Ital Biol* 140: 315–322, 2002.

Angelaki DE and Dickman JD. Spatiotemporal processing of linear acceleration: Primary afferent and central vestibular neuron responses. *J Neurophysiol* 84: 2113–2132, 2000.

Angelaki DE and Dickman JD. Neural encoding of gaze dependencies during translation. In: *Levels of Perception*, edited by Harris LR and Jerkin M. Berlin: Springer Verlag, 2002, p. 319–340.

Angelaki DE and Hess BJM. Direction of heading and vestibular control of binocular eye movements. *Vision Res* 41: 3215–3228, 2001.

Angelaki DE and McHenry MQ. Short-latency primate vestibulo-ocular responses during translation. *J Neurophysiol* 82: 651–1654, 1999.

- Angelaki DE, Green AM, and Dickman JD. Differential sensorimotor processing of vestibulo-ocular signals during rotation and translation. *J Neurosci* 21: 3968–3985, 2001.
- Angelaki DE, McHenry MQ, Dickman JD, Newlands SD, and Hess BJM. Computation of inertial motion: neural strategies to resolve ambiguous otolith information. *J Neurosci* 19: 316–327, 1999.
- Angelaki DE, McHenry MQ, Dickman JD, and Perachio AA. Primate translational vestibuloocular reflexes. III. Effects of bilateral labyrinthine electrical stimulation. *J Neurophysiol* 83: 1662–1676, 2000a.
- Angelaki DE, McHenry MQ, and Hess BJM. Primate translational vestibuloocular reflexes. I High frequency dynamics and three-dimensional properties during lateral motion. *J Neurophysiol* 83: 1637–1647, 2000b.
- Angelaki DE, Newlands SD, and Dickman JD. Primate translational vestibuloocular reflexes. IV. Changes after unilateral labyrinthectomy. *J Neurophysiol* 83: 3005–3018, 2000c.
- Angelaki DE, Shaikh A, and Dickman JD. Responses of vestibular nuclei neurons during combinations of tilt and translation. Program No. 593.8 2003 Abstract Viewer/Itinerary Planner. Washington DC: Society for Neuroscience, 2003. Online.
- Angelaki DE, Zhou HH, and Wei M. Foveal versus full-field visual stabilization strategies for translational and rotational head movements. *J Neurosci* 23: 1104–1108, 2003.
- Aw ST, Todd MJ, McGarvie LA, Migliaccio AA, and Halmagyi GM. Effects of unilateral vestibular deafferentation on the linear vestibuloocular reflex evoked by impulsive eccentric roll rotation. *J Neurophysiol* 89: 969–978, 2003.
- Baarsma EA and Collewijn H. Eye movements due to linear accelerations in the rabbit. *J Physiol* 245: 227–247, 1975.
- Bai R, Meng H, Sato H, Imagawa M, Sasaki M, and Uchino Y. Properties of utricular-activated vestibular neurons that project to the contralateral vestibular nuclei in the cat. *Exp Brain Res* 147: 419–425, 2002.
- Baker R, Mano N, and Shimazu H. Postsynaptic potentials in abducens motoneurons induced by vestibular stimulation. *Brain Res* 15: 577–580, 1969.
- Baker R, Precht W, and Llinás R. Cerebellar modulatory action on the vestibulo-trochlear pathway in the cat. *Exp Brain Res* 15: 364–385, 1972.
- Baloh RW, Beykirch K, Honrubia V, and Yee RD. Eye movements induced by linear acceleration on a parallel swing. *J Neurophysiol* 60: 2000–2013, 1988.
- Baloh RW, Yue Q, Jacobson KM, and Honrubia V. Persistent direction-changing positional nystagmus: another variant of benign positional nystagmus? *Neurology* 45: 1297–1301, 1995.
- Bisdorff AR, Bronstein AM, Wolsley C, Gresty MA, Davies A, and Young A. EMG responses to free fall in elderly subjects and akinetic rigid patients. *J Neurol Neurosurg Psychiatry* 66: 447–455, 1999.
- Bloomberg JJ, Reschke MF, Huebner WP, and Peters BT. The Effects of target distance on eye and head movement during locomotion. *Ann NY Acad Sci* 656: 699–707, 1992.
- Bolton PS, Endo K, Goto M, Imagawa M, Sasaki M, Uchino Y, and Wilson VJ. Connections between utricular nerve and dorsal neck motoneurons of the decerebrate cat. *J Neurophysiol* 67: 1695–1697, 1992.
- Bronstein AM and Gresty MA. Short latency compensatory eye movement responses to transient linear head acceleration: a specific function of the otolith-ocular reflex. *Exp Brain Res* 71: 406–410, 1988.
- Bronstein AM, Gresty MA, and Brookes GB. Compensatory otolith slow phase eye movement responses to abrupt linear head motion in the lateral direction. Finding in patients with labyrinthine and neurological lesions. *Acta Otolaryngol Suppl* 481: 42–46, 1991.
- Buizza A, Léger A, Droulez J, Berthoz A, and Schmid R. Influence of otolith stimulation by horizontal linear acceleration on optokinetic nystagmus and visual motion perception. *Exp Brain Res* 39: 165–176, 1980.
- Busettoni C, Masson GS, and Miles FA. A role for stereoscopic depth cues in the rapid visual stabilization of the eyes. *Nature* 380: 342–345, 1996a.
- Busettoni C, Masson GS, and Miles FA. Radial optic flow induces vergence eye movements with ultra-short latencies. *Nature* 390: 512–515, 1997.
- Busettoni C, Miles FA, and Krauzlis RJ. Short-latency disparity vergence responses and their dependence on a prior saccadic eye movement. *J Neurophysiol* 75: 1392–1410, 1996b.
- Busettoni C, Miles FA, Schwarz U, and Carl JR. Human ocular responses to translation of the observer and of the scene: dependence on viewing distance. *Exp Brain Res* 100: 484–494, 1994.
- Bush GA and Miles FA. Short-latency compensatory eye movements associated with a brief period of free fall. *Exp Brain Res* 108: 337–340, 1996.
- Chen-Huang C and McCrea RA. Contribution of vestibular nerve irregular afferents to viewing distance-related changes in the vestibulo-ocular reflex. *Exp Brain Res* 119: 116–130, 1998a.
- Chen-Huang C and McCrea RA. Viewing distance related sensory processing in the ascending tract of deiters vestibulo-ocular reflex pathway. *J Vestib Res* 8: 175–184, 1998b.
- Chen-Huang C and McCrea RA. Effects of viewing distance on the responses of vestibular neurons to combined angular and linear vestibular stimulation. *J Neurophysiol* 81: 2538–2557, 1999a.
- Chen-Huang C and McCrea RA. Effects of viewing distance on the responses of horizontal canal-related secondary vestibular neurons during angular head rotation. *J Neurophysiol* 81: 2517–2537, 1999b.
- Chubb MC, Fuchs AF, and Scudder C. A. Neuron activity in monkey vestibular nuclei during vertical vestibular stimulation and eye movements. *J Neurophysiol* 52: 724–742, 1984.
- Clark B and Graybiel A. Factors contributing to the delay in the perception of the oculogravic illusion. *Am J Physiol* 79: 377–388, 1966.
- Collewijn H, Van Der Steen J, Ferman L and Jansen TC. Human ocular counterroll: assessment of static and dynamic properties from electromagnetic scleral coil recordings. *Exp Brain Res* 59: 185–196, 1985.
- Crane BT and Demer JL. Human gaze stabilization during natural activities: translation, rotation, magnification and target distance effects. *J Neurophysiol* 78: 2129–2144, 1997.
- Crane BT and Demer JL. Human horizontal vestibulo-ocular reflex initiation: effects of acceleration, target distance and unilateral deafferentation. *J Neurophysiol* 80: 1151–1166, 1998.
- Crane BT, Tian J-R, and Demer JL. Initial vestibulo-ocular reflex during transient angular and linear acceleration in human cerebellar dysfunction. *Exp Brain Res* 130: 486–496, 2000.
- Crane BT, Tian J, Wiest G, and Demer JL. Initiation of the human heave linear vestibulo-ocular reflex. *Exp Brain Res* 148: 247–255, 2003.
- Crawford JD, Tweed DB, and Vilis T. Static ocular counterroll is implemented through the 3-D neural integrator. *J Neurophysiol* 90: 2777–2784, 2003.
- Cullen KE and McCrea RA. Firing behaviour of brain stem neurons during voluntary cancellation of the horizontal vestibuloocular reflex. I. Secondary vestibular neurons. *J Neurophysiol* 70: 828–843, 1993.
- Demer JL and Viirre E. Visual-vestibular interaction during standing, walking, and running. *J Vest Res* 6: 295–313, 1996.
- Diamond SG, Markham CH, Simpson NE and Curthoys IS. Binocular counterrolling in humans during dynamic rotation. *Acta Otolaryngol* 87: 490–498 1979.
- Dickman JD and Angelaki DE. Three-dimensional organization of vestibular related eye movements during off-vertical axis rotation and linear translation in pigeons. *Exp Brain Res* 129: 391–400, 1999.
- Dickman JD and Angelaki DE. Vestibular convergence patterns in vestibular nuclei neurons of alert primates. *J Neurophysiol* 88: 3518–3533, 2002.
- Einstein A. Über das Relativitätsprinzip und die aus demselben gezogenen Folgerungen. *Jahrb Radioakt* 4: 411–462, 1908.
- Fernandez C and Goldberg JM. Physiology of peripheral neurons innervating semicircular canals of the squirrel monkey. II. Response to sinusoidal stimulation and dynamics of peripheral vestibular system. *J Neurophysiol* 34: 661–675, 1971.
- Fernandez C and Goldberg JM. Physiology of peripheral neurons innervating otolith organs of the squirrel monkey. I. Response to static tilts and to long-duration centrifugal forces. *J Neurophysiol* 39: 970–984, 1976a.
- Fernandez C and Goldberg JM. Physiology of peripheral neurons innervating otolith organs of the squirrel monkey. II. Directional selectivity and force-response relations. *J Neurophysiol* 39: 985–995, 1976b.
- Fernandez C and Goldberg JM. Physiology of peripheral neurons innervating otolith organs of the squirrel monkey: III. Response dynamics. *J Neurophysiol* 39: 996–1008, 1976c.
- Flock A. Structure of the macula utriculi with special reference to directional interplay of sensory responses as revealed by morphological polarization. *J Cell Biol* 22: 413–431, 1964.
- Fuchs AF and Kimm J. Unit activity in vestibular nucleus of the alert monkey during horizontal angular accelerations and eye movements. *J Neurophysiol* 38: 1140–1161, 1975.
- Gianna CC, Gresty MA, and Bronstein AM. Influence of target distance and acceleration level on eye movements evoke by lateral acceleration steps. *Acta Otolaryngol* S520: 65–67, 1995.
- Gianna CC, Gresty MA, and Bronstein AM. Eye movements induced by lateral acceleration steps: effect of visual context and acceleration levels. *Exp Brain Res* 114: 124–129, 1997.

- Gianna CC, Gresty MA, and Bronstein AM.** The human linear vestibulo-ocular reflex to transient accelerations: visual modulation of suppression and enhancement. *J Vest Res* 10: 227–238, 2000.
- Glasauer S and Merfeld DM.** Modelling three-dimensional vestibular responses during complex motion stimulation. In: *Three-Dimensional Kinematics of Eye, Head and Limb Movements*, edited by Fetter M, Haslwanter T, Misslisch H, and Tweed D. Amsterdam: Harwood Academic, 1997, p. 387–398.
- Glasauer S, Dieterich M and Brandt T.** Central positional nystagmus simulated by a mathematical ocular motor model of otolith-dependent modification of Listing's plane. *J Neurophysiol* 86: 1546–1554, 2001.
- Gomi H, Shidara M, Takemura A, Inoue Y, Kawano K, and Kawato M.** Temporal firing patterns of Purkinje cells in the cerebellar ventral paraflocculus during ocular following responses in monkeys. I. Simple spikes. *J Neurophysiol* 80: 818–831, 1998.
- Green AM and Galiana HL.** Hypothesis for shared central processing of canal and otolith signals. *J Neurophysiol* 80: 2222–2228, 1998.
- Green AM and Angelaki DE.** An integrative neural network for detecting head orientation and movement. *J Neurophysiol* In press.
- Green AM and Angelaki DE.** Resolution of sensory ambiguities for gaze stabilization requires a second neural integrator. *J Neurosci* 23: 9265–9275, 2003.
- Guedry FE Jr.** *Handbook of Sensory Physiology—Vestibular System—Psychophysics, Applied Aspects and General Interpretations*, edited by Guedry Jr FE. Berlin: Springer-Verlag, 1974, part 2, p. 1–154.
- Hannan RA, Kabrisky M, Replogle CR, Hartzler VL, and Roccaforte PA.** Experimental determination of a portion of the human vestibular system response through measurement of eyeball counterroll. *IEEE Trans Bio-Med Eng* 13: 65–70, 1966.
- Haslwanter T, Straumann D, Hepp K, Hess BJM, and Henn V.** Smooth pursuit eye movements obey Listing's law in the monkey. *Exp Brain Res* 87: 470–472, 1991.
- Haslwanter T, Straumann D, Hess BJM, and Henn V.** Static roll and pitch in the monkey: shift and rotation of Listing's plane. *Vision Res* 32: 1341–1348, 1992.
- Hess BJM and Angelaki DE.** Oculomotor control of primary eye position discriminates between translation and tilt. *J Neurophysiol* 81: 394–398, 1999.
- Hess BJM and Angelaki DE.** Gravity modulates Listing's plane orientation during both pursuit and saccades. *J Neurophysiol* 90: 1340–1345, 2003a.
- Hess BJM and Angelaki DE.** Vestibular contributions to gaze stability during transient forward and backward motion. *J Neurophysiol* 90: 1996–2004, 2003b.
- Hess BJM and Dieringer N.** Spatial organization of the maculo-ocular reflex of the rat: responses during off-vertical axis rotation. *Eur J Neurosci* 2: 909–919, 1990.
- Hess BJM and Dieringer N.** Spatial organization of linear vestibuloocular reflexes of the rat: responses during horizontal and vertical linear acceleration. *J Neurophysiol* 66: 1805–1818, 1991.
- Hess BJM, Knöpfel T, and Precht W.** Dynamics of maculo-ocular reflexes in the frog. *Neuroscience* 11: 645–650, 1984.
- Highstein SM.** Synaptic linkage in the vestibulo-ocular and cerebello-vestibular pathways to the VIth nucleus in the rabbit. *Exp Brain Res* 17: 301–314, 1973.
- Hine T and Thorn F.** Compensatory eye movements during active head rotation for near targets: effects of imagination, rapid head oscillation and vergence. *Vision Res* 9: 1639–1657, 1987.
- Huterer M and Cullen KE.** Vestibuloocular reflex dynamics during high-frequency and high-acceleration rotations of the head on body in rhesus monkey. *J Neurophysiol* 88: 13–28, 2002.
- Ikegami H, Sasaki M, and Uchino Y.** Connections between utricular nerve and neck flexor motoneurons of decerebrate cats. *Exp Brain Res* 98: 373–378, 1994.
- Imagawa M, Isu N, Sasaki M, Endo K, Ikegami H, and Uchino Y.** Axonal projections of utricular afferents to the vestibular nuclei and the abducens nucleus in cats. *Neurosci Lett* 186: 87–90, 1995.
- Israël I and Berthoz A.** Contribution of the otoliths to the calculation of linear displacement. *J Neurophysiol* 62: 247–263, 1989.
- Ito M, Nisimaru N, and Yamamoto M.** Specific patterns of neuronal connections involved in the control of the rabbit's vestibulo-ocular reflexes by the cerebellar flocculus. *J Physiol* 265: 833–854, 1977.
- Ito Y, Corna S, von Brevern M, Bronstein A, and Gresty M.** The functional effectiveness of neck muscle reflexes for head-righting in response to sudden fall. *Exp Brain Res* 117: 266–272, 1997.
- Kawano K.** Ocular tracking behavior and neurophysiology. *Curr Opin Neurobiol* 9: 467–473, 1999.
- Kawano K and Shidara M.** The neural pathway that mediates ocular following responses. *Biomed Res* 14: 81–83, 1993.
- Keller EL and Daniels PD.** Oculomotor related interaction of vestibular and visual stimulation in vestibular nucleus cells in alert monkey. *Exp Neurol* 46: 187–198, 1975.
- Keller EL and Kamath BY.** Characteristics of head rotation and eye movement-related neurons in alert monkey vestibular nucleus. *Brain Res* 100: 182–187, 1975.
- Kellogg RS.** Dynamic counterrolling of the eye in normal subjects and in persons with bilateral labyrinthine defects. *NASA S-77* 195–202, 1965.
- King WM, Lisberger SG, and Fuchs AF.** Responses of fibers in medial longitudinal fasciculus (MLF) of alert monkeys during horizontal and vertical conjugate eye movements evoked by vestibular or visual stimuli. *J Neurophysiol* 39: 1135–1149, 1976.
- King W, Tang B, and Zhou W.** Translational vestibulo-ocular reflex (TVOR) signal preprocessing by monkey vestibular neurons. *Abstr Soci Neurosci* 593.21, 2003.
- Kobayashi Y, Kawano K, Takemura A, Inoue Y, Kitama T, Gomi H, and Kawato M.** Temporal firing patterns of Purkinje cells in the cerebellar ventral paraflocculus during ocular following responses in monkeys. II. Complex spikes. *J Neurophysiol* 80: 832–848, 1998.
- Krejcová H, Highstein SM, and Cohen B.** Labyrinthine and extra-labyrinthine effects on ocular counter-rolling. *Acta Otolaryngol* 72: 165–171, 1971.
- Kushiro K, Zakir M, Ogawa Y, Sato H, and Uchino Y.** Saccular and utricular inputs to sternocleidomastoid motoneurons of decerebrate cats. *Exp Brain Res* 126: 410–416, 1999.
- Kushiro K, Zakir M, Sato H, Ono S, Ogawa Y, Meng H, Zhang X, and Uchino Y.** Saccular and utricular inputs to single vestibular neurons in cats. *Exp Brain Res* 131: 406–415, 2000.
- Langer TP, Fuchs AF, Chubb MC, Scudder CA, and Lisberger SG.** Floccular efferents in the rhesus macaque as revealed by autoradiography and horseradish peroxidase. *J Comp Neur* 235: 26–37, 1985a.
- Langer TP, Fuchs AF, Scudder CA, and Chubb MC.** Afferents to the flocculus of the cerebellum in the rhesus macaque as revealed by retrograde transport of horseradish peroxidase. *J Comp Neurol* 235: 1–25, 1985b.
- Lempert T, Gianna C, Brookes G, Bronstein A, and Gresty M.** Horizontal otolith-ocular responses in humans after unilateral vestibular deafferentation. *Exp Brain Res* 118: 533–540, 1998.
- Lichtenberg BK, Young LR, and Arrott AP.** Human ocular counterrolling induced by varying linear accelerations. *Exp Brain Res* 48: 127–136, 1982.
- Lindeman HH.** *Studies on the Morphology of the Sensory Regions of the Vestibular Apparatus.* Ergebnisse Der Anatomie Und Entwicklungs-geschichte 62: 1–113, 1969.
- Lisberger SG and Pavelko TA.** Brain stem neurons in modified pathways for motor learning in the primate vestibulo-ocular reflex. *Science* 242: 771–773, 1988.
- Lisberger SG, Pavelko TA, and Broussard DM.** Responses during eye movements of brain stem neurons that receive monosynaptic inhibition from the flocculus and ventral paraflocculus in monkeys. *J Neurophysiol* 72: 909–927, 1994.
- Maruta J, Simpson JI, Raphan T, and Cohen B.** Orienting otolith-ocular reflexes in the rabbit during static and dynamic tilts and off-vertical axis rotation. *Vision Res* 41: 3255–3270, 2001.
- Mayne R.** A systems concept of the vestibular organs. *Handbook of Sensory Physiology: Vestibular System*, edited by Kornhuber HH. New York: Springer-Verlag, 1974, p. 493–580.
- McCrea RA, Strassman A, May E, and Highstein SM.** Anatomical and physiological characteristics of vestibular neurons mediating the horizontal vestibulo-ocular reflex of the squirrel monkey. *J Comp Neurol* 264: 547–570, 1987.
- McFarland JL and Fuchs AF.** Discharge patterns in nucleus prepositus hypoglossi and adjacent medial vestibular nucleus during horizontal eye movement in behaving macaques. *J Neurophysiol* 68: 319–332, 1992.
- McConville K, Tomlinson RD, and Na E-Q.** Behavior of eye-movement-related cells in the vestibular nuclei during combined rotational and translation stimuli. *J Neurophysiol* 76: 3136–3148, 1996.
- McHenry MQ and Angelaki DE.** Primate translational vestibulo-ocular reflexes. II. Response and vergence responses to fore-aft motion. *J Neurophysiol* 83: 1648–1661, 2000.
- Mederndorp WP, Van Gisbergen JAM, and Gielen CCAM.** Human gaze stabilization during active head translations. *J Neurophysiol* 87: 295–304, 2002.

- Melville Jones G, Rolph R, and Downing GH. Comparison of human subjective and oculomotor responses to sinusoidal vertical linear acceleration. *Acta Otolaryngol* 90: 431–440, 1980.
- Meng H and Angelaki DE. Viewing distance modulation of premotor vestibular neurons. *Soc Neurosci Abstr* 391.9, 2003.
- Merfeld DM. Modeling the vestibulo-ocular reflex of the squirrel monkey using eccentric rotation and roll tilt. *Exp Brain Res* 106: 123–134, 1995.
- Merfeld DM, Park S, Gianna-Poulin C, Black F, and Wood S. Neural strategies to resolve ambiguous otolith cues: influence of rotational cues on human perception of tilt and translation. *Soc Neurosci Abstr*, 268.18, 2003.
- Merfeld DM and Zupan LH. Neural processing of gravito-inertial cues in humans. III. Modeling tilt and translation responses. *J Neurophysiol* 87: 819–833, 2002.
- Merfeld DM, Zupan LH, and Gifford CA. Neural processing of gravito-inertial cues in humans. II. Influence of the semicircular canals during eccentric rotation. *J Neurophysiol* 85: 1648–1660, 2001.
- Merfeld DM, Zupan L, and Peterka RJ. Humans use internal models to estimate gravity and linear acceleration. *Nature* 398: 615–618, 1999.
- Mergner T and Glasauer S. A simple model of vestibular canal-otolith signal fusion. *Ann NY Acad Sci* 28: 430–434, 1999.
- Miles FA. Single unit firing patterns in the vestibular nuclei related to voluntary eye movements and passive body rotation in conscious monkeys. *Brain Res* 71: 215–224, 1974.
- Miles FA. The sensing of rotational and translational optic flow by the primate optokinetic system. In: *Visual Motion and its Role in the Stabilization of Gaze* edited by Miles FA and Wallman, J. Amsterdam: Elsevier, 1993, p. 393–403.
- Miles FA. The sensing of optic flow by the primate optokinetic system. In: *Eye Movement Research: Mechanisms, Processes and Applications*, edited by Findlay JM, Kentridge RW, and Walker R. Amsterdam: Elsevier, 1995, p. 47–57.
- Miles FA. Visual stabilization of the eyes in primates. *Curr Opin Neurobiol* 7: 867–871, 1997.
- Miles FA. The neural processing of 3-D visual information: evidence from eye movements. *Eur J Neurosci* 10: 811–822, 1998.
- Miles FA and Busetini C. Ocular compensation for self motion: visual mechanisms. In: *Sensing and Controlling Motion: Vestibular and Sensorimotor Function*, edited by Cohen B, Tomko DL, and Guedry F. New York: Ann NY Acad Sci, 1992, p. 220–232.
- Miles FA, Schwarz U, and Busetini C. The parsing of optic flow by the primate oculomotor system. In: *Representations of Vision: Trends and Tacit Assumptions in Vision Research*, edited by A Gorea A. Cambridge, MA: Cambridge Univ. Press, 1991, p. 185–199.
- Minor LB and Goldberg JM. Vestibular-nerve inputs to the vestibulo-ocular reflex: a functional-ablation study in the squirrel monkey. *J Neurosci* 11: 1636–1648, 1991.
- Minor LB, Lasker DM, Backous DD, and Hullar TE. Horizontal vestibulo-ocular reflex evoked by high-acceleration rotations in the squirrel monkey. I. Normal responses. *J Neurophysiol* 82: 1254–1270, 1999.
- Misslisch H and Hess BJM. Three-dimensional vestibuloocular reflex of the monkey: optimal retinal image stabilization versus Listing's law. *J Neurophysiol* 83: 3264–3276, 2000.
- Misslisch H, Tweed D, Sievering D, and Koenig E. Rotational kinematics of the human vestibuloocular reflex. III. Listing's law. *J Neurophysiol* 72: 2490–2502, 1994.
- Moore ST, Hirasaki E, Cohen B, and Raphan T. Effect of viewing distance on the generation of vertical eye movements during locomotion. *Exp Brain Res* 129: 347–361, 1999.
- Mussallum WS and Tomlinson RD. Model for the translational vestibuloocular reflex (VOR). *J Neurophysiol* 82: 2010–2014, 1999.
- Nagao S, Kitamura T, Nakamura N, Hiramatsu T, and Yamada J. Differences of the primate flocculus and ventral paraflocculus in the mossy and climbing fiber input organization. *J Comp Neur* 382: 480–498, 1997.
- Newlands SD, Purcell IM, Kevetter GA, and Perachio AA. Central projection of the utricular nerve in the gerbil. *J Comp Neurol* 452: 11–23, 2002.
- Newlands SD, Vrabec JT, Purcell IM, Stewart CM, Zimmerman BE, and Perachio AA. Central projections of the saccular and utricular nerves in macaques. *J Comp Neurol* 466: 31–47, 2003.
- Niven JI, Hixson WC, and Correia MJ. Elicitation of horizontal nystagmus by periodic linear acceleration. *Acta Otolaryngol* 62: 429–441, 1966.
- Ogawa Y, Kushihiro K, Zakir M, Sato H, and Uchino Y. Neuronal organization of the utricular macula concerned with innervation of single vestibular neurons in the cat. *Neurosci Lett* 278: 89–92, 2000.
- Paige GD. The influence of target distance on eye movement responses during vertical linear motion. *Exp Brain Res* 77: 585–593, 1989.
- Paige GD and Tomko DL. Eye movement responses to linear head motion in the squirrel monkey. I. Basic characteristics. *J Neurophysiol* 65: 1170–1182, 1991a.
- Paige GD and Tomko DL. Eye movement responses to linear head motion in the squirrel monkey. II. Visual-vestibular interactions and kinematic considerations. *J Neurophysiol* 65: 1183–1196, 1991b.
- Paige GD, Telford L, Seidman SH, and Barnes GR. Human vestibuloocular reflex and its interactions with vision and fixation distance during linear and angular head movement. *J Neurophysiol* 80: 2391–2404, 1998.
- Park S, Merfeld DM, Gianna-Poulin C, Black F, and Wood S. Neural strategies to resolve ambiguous otolith cues: Influence of rotational cues on human eye movement. *Soc Neurosci Abstr* 593.18, 2003.
- Partsalis AM, Zhang Y, and Highstein SM. Dorsal Y group in the squirrel monkey. I. Neuronal responses during rapid and long-term modifications of the vertical VOR. *J Neurophysiol* 73: 615–631, 1995a.
- Partsalis AM, Zhang Y, and Highstein SM. Dorsal Y group in the squirrel monkey. II. Contribution of the cerebellar flocculus to neuronal responses in normal and adapted animals. *J Neurophysiol* 73: 632–650, 1995b.
- Precht W, Richter A, and Grippio J. Responses of neurones in cat's abducens nuclei to horizontal angular acceleration. *Pflügers Arch* 309: 285–309, 1969.
- Purcell IM and Perachio AA. Peripheral patterns of terminal innervation of vestibular primary afferent neurons projecting to the vestibulocerebellum in the gerbil. *J Comp Neurol* 433: 48–61, 2001.
- Ramat S and Zee DS. Ocular motor responses to abrupt interaural head translation in normal humans. *J Neurophysiol* 90: 887–902, 2003.
- Richter A and Precht W. Inhibition of abducens motoneurons by vestibular nerve stimulation. *Brain Res* 11: 701–705, 1968.
- Robinson DA. The use of control systems analysis in the neurophysiology of eye movements. *Annu Rev Neurosci* 4: 463–503, 1981.
- Sato Y, Kanda K-I, and Kawasaki T. Target neurons of floccular middle zone inhibition in medial vestibular nucleus. *Brain Res* 446: 225–235, 1988.
- Sato H, Endo K, Ikegami H, Imagawa M, Sasaki M, and Uchino Y. Properties of utricular nerve-activated vestibulospinal neurons in cats. *Exp Brain Res* 112: 197–202, 1996.
- Sato H, Imagawa M, Isu N, and Uchino Y. Properties of saccular nerve-activated vestibulospinal neurons in cats. *Exp Brain Res* 116: 381–388, 1997.
- Schor RH and Miller AD. Vestibular reflexes in neck and forelimb muscles evoked by roll tilt. *J Neurophysiol* 46: 167–178, 1981.
- Schwarz U, Busetini C, and Miles FA. Ocular responses to linear motion are inversely proportional to viewing distance. *Science* 245: 1394–1396, 1989.
- Schwarz U and Miles FA. Ocular responses to translation and their dependence on viewing distance. I. Motion of the observer. *J Neurophysiol* 66: 851–864, 1991.
- Schwindt PC, Richter A, and Precht W. Short latency utricular and canal input to ipsilateral abducens motoneurons. *Brain Res* 60: 259–262, 1973.
- Scudder CA and Fuchs AF. Physiological and behavioral identification of vestibular nucleus neurons mediating the horizontal vestibuloocular reflex in trained rhesus monkeys. *J Neurophysiol* 86: 244–264, 1992.
- Seidman SH, Paige GD, and Tomko DL. Adaptive plasticity in the naso-occipital linear vestibulo-ocular reflex. *Exp Brain Res* 125: 485–494, 1999.
- Seidman SH, Telford L, and Paige GD. Tilt perception during dynamic linear acceleration. *Exp Brain Res* 119: 307–314, 1998.
- Shidara M and Kawano K. Role of Purkinje cells in the ventral paraflocculus in short-latency ocular following responses. *Exp Brain Res* 93: 185–1295, 1993.
- Shimazu H and Smith CM. Cerebellar and labyrinthine influences on single vestibular neurons identified by natural stimuli. *J Neurophysiol* 34: 493–508, 1971.
- Siegborn J and Grant G. Brainstem projections of different branches of the vestibular nerve. An experimental study by transganglionic transport of horseradish peroxidase in the cat. I. The horizontal ampullar and utricular nerves. *Arch Ital Biol* 121: 237–248, 1983.
- Skavenski AA and Robinson DA. Role of abducens neurons in vestibuloocular reflex. *J Neurophysiol* 36: 724–738, 1973.
- Skipper JJ and Barnes GR. Eye movements induced by linear acceleration are modified by visualisation of imaginary targets. *Acta Otolaryngol* S468: 289–293, 1989.
- Snyder LH and King WM. Effect of viewing distance and location of the axis of head rotation on the monkey's vestibuloocular reflex. I. Eye-movement responses. *J Neurophysiol* 67: 861–874, 1992.

- Snyder LH and King WM. Behavior and physiology of the macaque vestibulo-ocular reflex response to sudden off-axis rotation: computing eye translation. *Brain Res Bulletin* 40: 293–302, 1996.
- Takemura A and Kawano K. Sensory-to-motor processing of the ocular following response. *Neurosci Res* 43: 201–206, 2002.
- Takemura A, Inoue Y, Kawano K, Quaia C, and Miles FA. Single-unit activity in cortical area MST associated with disparity-vergence eye movements: evidence for population coding. *J Neurophysiol* 85: 2245–2266, 2001b.
- Takemura A, Inoue Y, Gomi H, Kawato M, and Kawano K. Change in neuronal firing patterns in the process of motor command generation for the ocular following response. *J Neurophysiol* 86: 1750–1763, 2001a.
- Telford L, Seidman SH, and Paige GD. Dynamics of squirrel monkey linear vestibuloocular reflex and interactions with fixation distance. *J Neurophysiol* 78: 1775–1790, 1997.
- Telford L, Seidman SH, and Paige GD. Canal-otolith interactions in the squirrel monkey vestibulo-ocular reflex and the influence of fixation distance. *Exp Brain Res* 118: 115–125, 1998.
- Tomko DL, and Paige GD. Linear vestibuloocular reflex during motion along axes between nasooccipital and interaural. *Ann New York Acad Sci* 656: 233–241, 1992.
- Tomlinson RD and Robinson DA. Signals in vestibular nucleus mediating vertical eye movements in the monkey. *J Neurophysiol* 51: 1121–1135, 1984.
- Tweed D, Fetter M, Andreadaki S, Koenig E, and Dichgans J. Three-dimensional properties of human pursuit eye movements. *Vision Res* 32: 1225–1238, 1992.
- Tweed D and Vilis T. Implications of rotational kinematics for the oculomotor system in three dimensions. *J Neurophysiol* 58: 832–849, 1987.
- Tweed D and Vilis T. Geometric relations of eye position and velocity vectors during saccades. *Vision Res* 30: 111–127, 1990.
- Uchino Y, Ikegami H, Sasaki M, Endo K, Imagawa M, and Isu N. Monosynaptic and disynaptic connections in the utriculo-ocular reflex arc of the cat. *J Neurophysiol* 71: 950–958, 1994.
- Uchino Y, Sasaki M, Sato H, Imagawa M, Suwa H, and Isu N. Utriculoocular reflex arc of the cat. *J Neurophysiol* 76: 1896–1903, 1996.
- Uchino Y, Sasaki M, Sato H, I, Imagawa M, Suwa H, and Isu N. Utricular input to cat extraocular motoneurons. *Acta Otolaryng* 528: 44–48, 1997a.
- Uchino Y, Sato H, Sasaki M, Imagawa M, Ikegami H, Isu N, and Graf W. Sacculocollic reflex arcs in cats. *J Neurophysiol* 77: 3003–3012, 1997b.
- Uchino Y, Sato H, and Suwa H. Excitatory and inhibitory inputs from saccular afferents to single vestibular neurons in the cat. *J Neurophysiol* 78: 2186–2192, 1997c.
- Uchino Y, Sato H, Zakir M, Kushiro K, Imagawa M, Ogawa Y, Ono S, Meng H, Zhang X, Katsuta M, Isu N, and Wilson VJ. Commissural effects in the otolith system. *Exp Brain Res* 136: 421–430, 2001.
- Viirre E, Tweed D, Milner K, and Vilis T. A reexamination of the gain of the vestibuloocular reflex. *J Neurophysiol* 56: 439–450, 1986.
- Wei M and Angelaki DE. Does head rotation contribute to gaze stability during passive translation? *J Neurophysiol* 91: 1913–1918, 2004.
- Wilson VJ, Boyle R, Fukushima K, Rose PK, Shinoda Y, Sugiuchi Y, and Uchino Y. The vestibulocollic reflex. *J Vest Res* 5: 147–170, 1995.
- Yang D-S, FitzGibbon EJ, and Miles FA. Short-latency vergence eye movements induced by radial optic flow in humans: dependence on ambient vergence level. *J Neurophysiol* 81: 945–949, 1999.
- Young LR. Perceptions of the body in space: mechanisms. In: *Handbook of Physiology. The Nervous System. Sensory Processes*. Oxford: Oxford Univ. Press, 1974, sect.1, part III, p. 1023–1066.
- Zee DS, Walker MF, and Ramat S. The cerebellar contribution to eye movements based upon lesions: binocular, three-axis control and the translational vestibulo-ocular reflex. *Ann NY Acad Sci* 956: 178–189, 2002.
- Zhang Y, Partsalis AM, and Highstein SM. Properties of superior vestibular nucleus neurons projecting to the cerebellar flocculus in the squirrel monkey. *J Neurophysiol* 69: 642–645, 1993.
- Zhang Y, Partsalis AM, and Highstein SM. Properties of superior vestibular nucleus flocculus target neurons in the squirrel monkey: II Signal components revealed by reversible flocculus inactivation. *J Neurophysiol* 73: 2279–2292, 1995a.
- Zhang Y, Partsalis AM, and Highstein SM. Properties of superior vestibular nucleus flocculus target neurons in the squirrel monkey. I General properties in comparison with flocculus projecting neurons. *J Neurophysiol* 73: 2261–2278, 1995b.
- Zhou HH, Wei M, and Angelaki DE. Motor scaling by viewing distance of early visual motion signals during smooth pursuit. *J Neurophysiol* 88: 2880–2885, 2002.
- Zhou W, Tang BF, and King WM. Vestibular neurons encode linear translation and head tilt with respect to gravity. *Soc Neurosci Abstr* 26: 1491, 2000.
- Zhou W, Tang BF, and King WM. Responses of rostral Fastigial neurons to linear acceleration in an alert monkey. *Exp Brain Res* 139: 111–115, 2001.
- Zhou W, Weldon P, Tang B, and King WM. Rapid motor learning in the translational vestibulo-ocular reflex. *J Neurosci* 23: 4288–4298, 2003.
- Zupan LH, Merfeld DM, and Darlot C. Using sensory weighting to model the influence of canal, otolith and visual cues on spatial orientation and eye movements. *Biol Cybern* 86: 209–230, 2002.
- Zupan LH, Peterka RJ, and Merfeld DM. Neural processing of gravito-inertial cues in humans. I. Influence of the semicircular canals following post-rotatory tilt. *J Neurophysiol* 84: 2001–2015, 2000.